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PRODUCT REQUIREMENT DOCUMENT (PRD)
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Project Name:
Civic Intelligence & Municipal Operations Platform (CIMOP)

Version:
1.0

Document Owner:
Product Team

Prepared By:
Principal Product Architect

Status:
Planning Phase

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======
1. PRODUCT OVERVIEW
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The Civic Intelligence & Municipal Operations Platform (CIMOP) is an AI-powered smart-city complaint management and municipal operations system designed to digitize, automate, monitor, and optimize civic issue reporting and resolution processes.

The platform enables citizens to report infrastructure and public service issues using text, images, videos, GPS coordinates, and voice recordings.

Artificial Intelligence automatically analyzes incoming complaints, categorizes them, determines severity levels, detects duplicates, identifies responsible departments, and prioritizes resolution workflows.

Municipal employees receive work assignments, update progress in real-time, and close complaints through structured workflows.

Administrators gain city-wide visibility through operational dashboards, analytics systems, GIS heatmaps, infrastructure intelligence modules, and predictive maintenance systems.

The platform is designed for:

• Municipal Corporations
• Smart Cities
• Government Departments
• Urban Local Bodies
• Universities
• Housing Societies

• Industrial Parks
• Special Economic Zones
• Township Management Authorities

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2. PROBLEM STATEMENT
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Current municipal complaint systems suffer from:

• Manual complaint classification
• Delayed issue assignment
• Lack of transparency
• Duplicate complaint overload
• Poor citizen communication
• No predictive infrastructure monitoring
• Limited accountability
• Fragmented department coordination
• Weak performance monitoring

As city populations grow, traditional complaint handling systems become inefficient and unable to scale.

The platform solves these challenges through automation and AI-driven operations.

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3. BUSINESS GOALS
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Primary Goals

1. Reduce complaint processing time.
2. Improve complaint resolution rates.
3. Increase citizen satisfaction.
4. Improve department accountability.
5. Reduce duplicate complaint volume.
6. Enable city-wide infrastructure intelligence.
7. Introduce predictive maintenance capabilities.
8. Create a unified municipal operations platform.

Success Metrics

• Complaint classification accuracy > 90%
• Duplicate detection accuracy > 85%
• Average resolution time reduction > 50%
• Citizen satisfaction score > 4.5/5
• Employee productivity increase > 30%
• Complaint assignment automation > 90%

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4. TARGET USERS

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USER TYPE 1: Citizen

Purpose:
Report civic issues and track progress.

Responsibilities:
• Submit complaints
• Upload evidence
• Track status
• Provide feedback

USER TYPE 2: Field Worker

Purpose:
Resolve assigned issues.

Responsibilities:
• Receive assignments
• View location
• Upload work evidence
• Update status

USER TYPE 3: Department Manager

Purpose:
Manage department operations.

Responsibilities:
• Review complaints
• Allocate work
• Monitor employees
• Escalate issues

USER TYPE 4: City Administrator

Purpose:
Monitor city-wide performance.

Responsibilities:
• Review analytics
• Monitor KPIs
• Manage departments
• Infrastructure planning

USER TYPE 5: Super Admin

Purpose:

Platform governance.

Responsibilities:
• User management
• Security management
• System configuration
• AI governance
• Platform administration

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5. CORE MODULES
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======

MODULE 1:
Citizen Engagement

MODULE 2:
Complaint Management

MODULE 3:
AI Complaint Analysis

MODULE 4:
Department Operations

MODULE 5:
Field Workforce Management

MODULE 6:
Notification Engine

MODULE 7:
GIS Intelligence

MODULE 8:
Analytics Platform

MODULE 9:
Predictive Maintenance

MODULE 10:
Administration System

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6. FUNCTIONAL REQUIREMENTS
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6.1 AUTHENTICATION

Features

• Citizen Registration

• Employee Registration
• Login
• Password Reset
• Email Verification
• Mobile Verification
• JWT Authentication
• OAuth Support

6.2 CITIZEN PROFILE

Features

• Personal Information
• Contact Information
• Complaint Statistics
• Complaint History
• Saved Locations
• Notification Preferences

6.3 COMPLAINT SUBMISSION

Input Methods

1. Text Complaint
2. Image Upload
3. Video Upload
4. Voice Complaint
5. GPS Location
6. Map Selection

Required Fields

• Complaint Title
• Description
• Category
• Location
• Evidence

Optional Fields

• Landmark
• Additional Notes

6.4 COMPLAINT TRACKING

Statuses

• Submitted
• AI Analyzing
• Assigned
• In Progress
• Verification Pending

• Resolved
• Closed
• Rejected

Features

• Timeline View
• Status Updates
• Assigned Department
• Resolution Evidence

6.5 AI ANALYSIS

Outputs

• Issue Category
• Priority Score
• Severity Score
• Responsible Department
• Duplicate Probability
• Fraud Probability

6.6 FIELD OPERATIONS

Features

• Task Assignment
• Route Navigation
• Evidence Upload
• Work Completion Reports
• Daily Work Queue

6.7 ESCALATION MANAGEMENT

Rules

Level 1:
Department Escalation

Level 2:
Regional Escalation

Level 3:
City Escalation

Automatic escalation triggers:

• SLA Breach
• Critical Severity
• Citizen Re-open Requests

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7. NON-FUNCTIONAL REQUIREMENTS
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Availability

99.9% uptime

Performance

API Response Time:
< 300 ms

AI Classification:
< 5 seconds

Image Analysis:
< 10 seconds

Complaint Submission:
< 2 seconds

Scalability

Support:

• 1 Million Citizens
• 100,000 Complaints Daily
• Thousands of Employees

Reliability

• Automatic Backups
• Disaster Recovery
• Multi-Zone Deployment

Security

• End-to-End Encryption
• RBAC
• Audit Logging
• AI Abuse Protection

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8. AI REQUIREMENTS
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AI SYSTEMS

1. Complaint Classification Engine

Categories

• Road

• Garbage
• Drainage
• Water
• Electricity
• Public Toilet
• Park
• Traffic
• Public Property
• Other

----

2. Computer Vision Engine

Detect

• Potholes
• Garbage
• Water Leakage
• Broken Street Lights
• Open Manholes
• Broken Benches
• Drainage Issues

----

3. Duplicate Detection Engine

Identify

• Same complaint
• Same location
• Same image
• Same infrastructure issue

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4. Fraud Detection Engine

Identify

• Spam
• Fake complaints
• Bot activity
• Duplicate accounts

----

5. Predictive Maintenance Engine

Predict

• Future potholes
• Garbage hotspots
• Water leak hotspots
• Infrastructure failures

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9. DOMAIN RESTRICTED AI ASSISTANT
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The AI Assistant MUST operate exclusively within the civic complaint domain.

Allowed Topics

• Complaint submission
• Complaint tracking
• Municipal services
• Department information
• Infrastructure issues
• Resolution process

Forbidden Topics

• Coding
• Programming
• Homework
• Mathematics
• Translation
• Story writing
• General knowledge
• Entertainment
• Career advice

Example

User:
Write Python code

Response:
I can only assist with civic complaints and municipal services.

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10. REPORTING & ANALYTICS
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Analytics Modules

• Complaint Trends
• Resolution Trends
• Department Efficiency
• Employee Productivity
• Infrastructure Health
• SLA Compliance
• Citizen Satisfaction

KPI Dashboard

• Average Resolution Time
• Complaints per Category

• Resolution Rate
• Escalation Rate
• Repeat Complaint Rate
• AI Accuracy
• Fraud Detection Rate

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11. FUTURE EXPANSION
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Phase 2

• IoT Integration
• Smart Sensors
• Traffic Cameras
• Water Monitoring Sensors

Phase 3

• Smart City Command Center
• Infrastructure Digital Twin
• Emergency Response AI

Phase 4

• Multi-City Federation
• State-Level Monitoring
• National Infrastructure Analytics

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12. SUCCESS CRITERIA
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Technical Success

• Stable platform
• High AI accuracy
• Scalable architecture

Operational Success

• Faster complaint resolution
• Improved employee efficiency

Citizen Success

• Increased transparency
• Better communication
• Higher satisfaction

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END OF DOCUMENT

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======
SYSTEM ARCHITECTURE DOCUMENT
Civic Intelligence & Municipal Operations Platform (CIMOP)
Version: 1.0
Architecture Type: Cloud-Native, AI-Driven, Event-Oriented Microservices
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TABLE OF CONTENTS

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======
1. ARCHITECTURE VISION
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======

The Civic Intelligence & Municipal Operations Platform (CIMOP) is designed
as a large-scale, cloud-native municipal intelligence platform capable of serving:

• 1,000,000+ Citizens
• 100,000+ Complaints Daily
• Thousands of Field Workers
• Hundreds of Departments
• Multiple Cities

The architecture must support:

• High Availability
• Horizontal Scalability
• AI Workloads
• Real-Time Communication
• GIS Operations

• Analytics Processing
• Secure Government Workflows
• Future IoT Expansion

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2. ARCHITECTURE PRINCIPLES
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PRINCIPLE 1
Service Isolation

Each major business capability exists as an independent service.

PRINCIPLE 2
API First

All communication occurs through APIs.

PRINCIPLE 3
AI as a Dedicated Layer

AI services are independent from business logic.

PRINCIPLE 4
Event-Driven Processing

Heavy workloads execute asynchronously.

PRINCIPLE 5
Horizontal Scalability

Every critical component can scale independently.

PRINCIPLE 6
Fault Isolation

Failure in one module must not impact others.

PRINCIPLE 7
Observability First

Every action is logged, traced, and monitored.

```
==========================================================================
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3. HIGH LEVEL SYSTEM ARCHITECTURE
==========================================================================
======

                        +-----------------+
                        |   Citizens      |
                        +--------+--------+
                                 |
                                 |
                        +--------v--------+
                        | Frontend Apps   |
                        +--------+--------+
                                 |
                                 |
                    +------------v------------+
                    |      API Gateway        |
                    +------------+------------+
                                 |
             +-------------------+------------------+
             |                   |                  |
             |                   |                  |
     +-------v------+   +--------v------+  +--------v------+
     | Auth Service|    |ComplaintSvc  |   | AI Gateway    |
     +--------------+   +---------------+  +---------------+
                                 |
                                 |
          +----------------------+----------------------+
          |                      |                      |
          |                      |                      |
 +--------v-------+   +----------v----------+  +--------v-------+
 | GIS Service    |   | Notification Svc    |  | Analytics Svc |
 +----------------+   +---------------------+  +----------------+

                                 |
                                 |
                      +----------v----------+
                      | Event Bus/RabbitMQ  |
                      +----------+----------+
                                 |
      +--------------------------+----------------------------+
      |                          |                            |
      |                          |                            |
+-----v-----+          +---------v------+           +---------v------+
| NLP Engine|          | Computer Vision|           | Fraud Engine   |
+------------+         +-----------------+          +-----------------+

                                 |
                                 |
                    +------------v------------+
                    | PostgreSQL Cluster      |
                    +------------+------------+
                                 |
                      +----------v----------+
                      | Redis Cache Layer   |
                      +---------------------+
```

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4. COMPONENT ARCHITECTURE
==
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FRONTEND LAYER

Components:

1. Citizen Web Portal
2. Citizen Mobile App
3. Employee Dashboard
4. Manager Dashboard
5. Administrator Dashboard
6. Super Admin Console

Responsibilities:

• UI Rendering
• Form Submission
• Map Interaction
• Notifications
• Authentication

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5. SERVICE ARCHITECTURE
==
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SERVICE 1
AUTHENTICATION SERVICE

Responsibilities

• Registration
• Login
• JWT Generation
• OAuth Login
• Password Reset
• Session Management

Database Access

Users Table
Roles Table

SERVICE 2
USER SERVICE

Responsibilities

• Citizen Profiles
• Employee Profiles
• Preferences

• Contact Information

SERVICE 3
COMPLAINT SERVICE

Responsibilities

• Complaint Creation
• Complaint Updates
• Complaint Tracking
• Complaint Search

Core Business Service

SERVICE 4
AI ANALYSIS SERVICE

Responsibilities

• NLP Classification
• Image Analysis
• Duplicate Detection
• Priority Scoring
• Fraud Scoring

SERVICE 5
ASSIGNMENT SERVICE

Responsibilities

• Department Assignment
• Worker Assignment
• Workload Distribution

SERVICE 6
NOTIFICATION SERVICE

Responsibilities

• Email
• SMS
• Push Notifications
• In-App Notifications

SERVICE 7
GIS SERVICE

Responsibilities

• Geocoding
• Reverse Geocoding
• Heatmaps
• Zone Detection

SERVICE 8
ANALYTICS SERVICE

Responsibilities

• KPI Generation
• Reports
• Dashboard Metrics

SERVICE 9
AUDIT SERVICE

Responsibilities

• User Activity Logs
• Security Logs
• Compliance Records

==

6. DATA FLOW ARCHITECTURE

==

Complaint Submission Flow

```
Citizen
   |
   v
Frontend
   |
   v
API Gateway
   |
   v
Complaint Service
   |
   v
Store Initial Complaint
   |
   v
Publish Event
   |
   v
RabbitMQ
   |
   v
AI Analysis Service
```

```
   |
   +--> NLP
   |
   +--> CV
   |
   +--> Fraud
   |
   +--> Duplicate Detection
   |
   v
Analysis Results
   |
   v
Assignment Service
   |
   v
Department Allocation
   |
   v
Notification Service
   |
   v
Citizen Updated
```

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======

7. AI PROCESSING ARCHITECTURE

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AI PIPELINE

INPUTS

1. Text
2. Voice
3. Image
4. Video

```
                Input
                  |
                  v

        +-----------------------+
        | Data Validation Layer|
        +-----------+-----------+
                  |
                  v

        +-----------------------+
        | Feature Extraction   |
        +-----------+-----------+
                  |
                  v

        +-----------------------+
        | NLP Classification   |
        +-----------+-----------+
```

```
                   |
                   v

         +-----------------------+
         | Image Analysis        |
         +-----------+-----------+
                     |
                     v

         +-----------------------+
         | Duplicate Detection   |
         +-----------+-----------+
                     |
                     v

         +-----------------------+
         | Fraud Detection       |
         +-----------+-----------+
                     |
                     v

         +-----------------------+
         | Priority Engine       |
         +-----------+-----------+
                     |
                     v

         Final Decision Output
```

==
======
8. COMPLAINT LIFECYCLE ARCHITECTURE
==
======

```
Draft
  |
Submitted
  |
AI Processing
  |
Assigned
  |
Accepted
  |
In Progress
  |
Verification Pending
  |
Resolved
  |
Citizen Feedback
  |
Closed
```

Escalation Path

Assigned

```
   |
SLA Violation
   |
Manager Escalation
   |
City Escalation
   |
Administrative Review
```

==
9. EVENT DRIVEN ARCHITECTURE
==

RabbitMQ Events

ComplaintCreated

Triggers:
AI Analysis

ComplaintAnalyzed

Triggers:
Assignment

ComplaintAssigned

Triggers:
Notifications

ComplaintResolved

Triggers:
Feedback Collection

FraudDetected

Triggers:
Security Review

EscalationTriggered

Triggers:
Manager Alerts

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10. STORAGE ARCHITECTURE
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PRIMARY DATABASE

PostgreSQL Cluster

Stores

• Users
• Complaints
• Assignments
• Departments
• Analytics Metadata

CACHE LAYER

Redis

Stores

• Sessions
• JWT Blacklists
• Frequently Accessed Complaints
• Dashboard Metrics

OBJECT STORAGE

MinIO / AWS S3

Stores

• Images
• Videos
• Audio Files
• Resolution Evidence

VECTOR DATABASE

Qdrant

Stores

• Complaint Embeddings
• Image Embeddings
• Duplicate Detection Vectors

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11. REAL TIME COMMUNICATION ARCHITECTURE
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Technology

WebSockets

Used For

• Complaint Status Updates
• Dashboard Updates
• Notifications

Push Notifications

Firebase Cloud Messaging

Used For

• Mobile Alerts
• Assignment Alerts

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12. GIS ARCHITECTURE
===
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Preferred Stack

Frontend

Mapbox

Backend

PostGIS

Capabilities

• Heatmaps
• Complaint Clustering
• Zone Detection
• Spatial Queries
• Infrastructure Mapping

GIS Flow

GPS
 |
Complaint
 |
PostGIS
 |
Spatial Analysis

|
Dashboard Visualization

==
13. ANALYTICS ARCHITECTURE
==

Operational Analytics

• Daily Complaints
• Department Performance
• Resolution Metrics

Strategic Analytics

• Infrastructure Health
• Hotspot Detection
• Recurring Issue Analysis

Predictive Analytics

• Future Potholes
• Future Garbage Hotspots
• Failure Forecasting

==
14. SCALABILITY ARCHITECTURE
==

Horizontal Scaling

Services independently scale.

Examples

10 API Instances

20 AI Workers

5 Notification Workers

3 GIS Workers

Load Balancing

Nginx

Load Distribution

Round Robin

Least Connections

==

15. HIGH AVAILABILITY ARCHITECTURE

==

Availability Goal

99.9%

Implementation

Multi-AZ Deployment

Redundant Nodes

Database Replication

Redis Replication

RabbitMQ Clustering

==

16. DISASTER RECOVERY ARCHITECTURE

==

Backups

Database:
Every 6 Hours

Media:
Daily

Vector Database:
Daily

Recovery Targets

RPO:
15 Minutes

RTO:
1 Hour

==

17. PERFORMANCE ARCHITECTURE

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Target Metrics

API Response:
<300 ms

Complaint Submission:
<2 seconds

AI Classification:
<5 seconds

Dashboard Load:
<2 seconds

Concurrent Users:
100,000+

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18. MULTI-TENANT ARCHITECTURE
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Future Ready

Tenant Examples

City A
City B
University Campus
Industrial Park

Each Tenant Has

• Departments
• Users
• Complaints
• Analytics

Tenant Isolation

tenant_id column strategy

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======
19. FUTURE ARCHITECTURE EVOLUTION
==
======

PHASE 2

IoT Sensors

Smart Cameras

Traffic Sensors

Water Sensors

PHASE 3

Digital Twin Platform

PHASE 4

National Civic Intelligence Network

PHASE 5

AI Autonomous Infrastructure Monitoring

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END OF DOCUMENT
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==
SECURITY & ACCESS CONTROL DOCUMENT
Civic Intelligence & Municipal Operations Platform (CIMOP)
Version: 1.0
Classification: Enterprise/Government Grade Security Architecture
==

TABLE OF CONTENTS

==
1. SECURITY OBJECTIVES
==

Primary Goals

1. Protect citizen data
2. Prevent unauthorized access
3. Protect municipal infrastructure data
4. Prevent AI misuse
5. Ensure accountability through audit trails
6. Secure complaint evidence files
7. Prevent spam and fraud
8. Maintain data integrity
9. Support government-grade compliance

Security Targets

• Zero Trust Architecture
• Least Privilege Access
• End-to-End Encryption
• Complete Auditability
• AI Abuse Protection
• Multi-Layer Defense

==
2. SECURITY PRINCIPLES
==

PRINCIPLE 1
Zero Trust

Never trust any request automatically.

Every request must be authenticated and authorized.

PRINCIPLE 2
Least Privilege

Users receive only the permissions required.

PRINCIPLE 3
Defense in Depth

Multiple security layers protect every resource.

PRINCIPLE 4
Secure by Default

Every feature starts locked down.

PRINCIPLE 5
Audit Everything

Every important action is recorded.

PRINCIPLE 6
AI Restriction First

AI can never become a general-purpose assistant.

==
3. THREAT MODEL
==

THREAT CATEGORY 1

Unauthorized Access

Examples

• Stolen passwords
• Credential stuffing
• Brute-force attacks

Mitigation

• MFA
• Rate limiting
• Account lockout
• Password hashing

THREAT CATEGORY 2

Data Leakage

Examples

• Database breach
• Storage exposure

Mitigation

• Encryption
• Access control
• Audit logging

THREAT CATEGORY 3

AI Abuse

Examples

• Asking coding questions
• Prompt injection
• Jailbreak attempts

Mitigation

• Intent classification
• Domain filtering
• Output validation

THREAT CATEGORY 4

Malicious File Uploads

Examples

• Malware
• Executables
• Scripts

Mitigation

• Virus scanning
• MIME validation
• File restrictions

THREAT CATEGORY 5

API Abuse

Examples

• Scraping
• DDoS
• Automation attacks

Mitigation

• Rate limits
• API Gateway
• WAF

==
======
4. IDENTITY & ACCESS MANAGEMENT (IAM)
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======

Identity Components

• User Identity
• Role Assignment
• Permission Assignment
• Session Identity
• Device Identity

Identity Flow

```
User
 |
Login
 |
Authentication
 |
Role Resolution
 |
Permission Resolution
 |
Access Token
 |
Resource Access
```

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5. AUTHENTICATION ARCHITECTURE
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Supported Methods

1. Email + Password
2. Mobile OTP
3. Google OAuth
4. Microsoft OAuth
5. Government SSO (Future)

Password Policy

Minimum Length: 12

Must Contain

• Uppercase
• Lowercase
• Number
• Special Character

Password Hashing

Algorithm

Argon2id

Reason

Best protection against modern cracking attacks.

Never Store

• Plain Passwords
• Reversible Passwords

==

6. AUTHORIZATION ARCHITECTURE (RBAC)

==

Authorization Model

RBAC + Permission Based Access

Structure

```
User
 |
Role
 |
Permissions
 |
Resources
```

Example

```
Citizen
 |
View Own Complaint

Citizen
 |
Create Complaint

Citizen
 |
Edit Profile
```

==

7. USER ROLES & PERMISSIONS

==

ROLE 1
Citizen

Permissions

✓ Create Complaint
✓ View Own Complaint
✓ Update Profile
✓ Submit Feedback
✓ Escalate Complaint

Cannot

✗ Assign Complaints
✗ View Others' Complaints
✗ Access Analytics

ROLE 2
Field Worker

Permissions

✓ View Assigned Tasks
✓ Update Complaint Status
✓ Upload Resolution Evidence

Cannot

✗ Delete Complaints
✗ Access City Analytics

ROLE 3
Department Manager

Permissions

✓ Assign Workers
✓ Reassign Tasks
✓ View Department Reports
✓ Escalate Complaints

ROLE 4
City Administrator

Permissions

✓ View All Complaints

✓ Access Analytics
✓ Manage Departments
✓ Manage Employees

ROLE 5
Super Admin

Permissions

✓ Full Platform Access
✓ Security Controls
✓ Role Management
✓ System Configuration
✓ AI Configuration

==
8. JWT SECURITY ARCHITECTURE
==

Token Types

1. Access Token
2. Refresh Token

Access Token

Lifetime

15 Minutes

Contains

• User ID
• Role
• Tenant ID
• Permissions

Refresh Token

Lifetime

30 Days

Stored

HTTP Only Cookie

Security Measures

✓ Token Rotation
✓ Token Revocation
✓ Redis Blacklist

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9. OAUTH ARCHITECTURE
==
======

Supported Providers

• Google
• Microsoft

Flow

```
User
 |
OAuth Provider
 |
Token Verification
 |
User Mapping
 |
JWT Generation
```

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10. SESSION MANAGEMENT
==
======

Session Store

Redis

Stored Data

• Active Sessions
• Refresh Tokens
• Device Metadata

Session Security

✓ Device Tracking
✓ Login History
✓ Session Revocation

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11. API SECURITY

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Security Layers

Layer 1
TLS Encryption

Layer 2
JWT Validation

Layer 3
Permission Validation

Layer 4
Rate Limiting

Layer 5
Input Validation

Layer 6
Audit Logging

Rate Limits

Citizen

100 Requests / Minute

Employee

300 Requests / Minute

Admin

500 Requests / Minute

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12. FILE UPLOAD SECURITY
==
======

Allowed Types

Images

- JPG
- JPEG
- PNG
- WEBP

Videos

• MP4

Audio

• WAV
• MP3

Maximum Size

Images:
20 MB

Videos:
100 MB

Audio:
20 MB

Security Checks

✓ MIME Validation
✓ Extension Validation
✓ Virus Scanning
✓ Metadata Inspection

Blocked Types

✗ EXE
✗ DLL
✗ BAT
✗ JS
✗ SH
✗ ZIP

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13. AI SECURITY & DOMAIN RESTRICTION
==
======

CRITICAL REQUIREMENT

AI MUST ONLY OPERATE INSIDE CIVIC DOMAIN.

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LAYER 1
INTENT CLASSIFIER

Purpose

Detect user intent.

Allowed

• Complaint Creation
• Complaint Tracking
• Civic Services

Blocked

• Coding
• Homework
• Math
• Entertainment

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LAYER 2
DOMAIN RELEVANCE DETECTOR

Checks

Does request belong to municipal domain?

Examples

Accepted

"Street light is broken."

Rejected

"Write Python code."

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LAYER 3
PROMPT GUARDRAILS

System Prompt Rules

AI must never answer:

• Programming
• Career Advice
• General Knowledge
• Translation

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LAYER 4
OUTPUT VALIDATION

Validate generated response.

If response contains:

• Code
• Educational Content
• Non-civic Answers

Response blocked.

==

LAYER 5
MODERATION LAYER

Detect

• Abuse
• Hate Speech
• Threats
• Illegal Content

==
14. DATA SECURITY
==

Data Classification

Public

Examples

Department names

Internal

Complaint metadata

Confidential

Citizen Information

Restricted

Audit Logs
Security Logs

==

15. ENCRYPTION ARCHITECTURE
==

Encryption In Transit

TLS 1.3

Encryption At Rest

AES-256

Encrypted Resources

• Database
• File Storage
• Backups
• Audit Logs

==
16. AUDIT LOGGING
==

Every Sensitive Action Logged

Examples

• Login
• Logout
• Complaint Creation
• Assignment
• Status Change
• Role Change
• Permission Change

Stored Fields

• User ID
• Timestamp
• Action
• Resource
• IP Address
• Device Info

==
17. FRAUD PREVENTION
==

Detect

• Spam Complaints
• Fake Complaints
• Duplicate Accounts
• Bot Activity

Signals

• Complaint Frequency
• Device Fingerprint
• IP Patterns
• Duplicate Media

Risk Score

0–100

Thresholds

0–30 Safe

31–60 Review

61–100 High Risk

==
18. ABUSE PREVENTION
==

Controls

✓ CAPTCHA
✓ Rate Limiting
✓ Device Fingerprinting
✓ Account Verification
✓ AI Abuse Detection

==
19. INFRASTRUCTURE SECURITY
==

Components

• Firewall
• WAF
• Kubernetes Network Policies
• Private Subnets
• Bastion Hosts

Public Access

Only

• Load Balancer

• API Gateway

==
20. DATABASE SECURITY
==

Controls

✓ Encryption
✓ Read Replicas
✓ Least Privilege Access
✓ Database Auditing

Restrictions

Application Users Cannot

✗ Access Raw Database
✗ Execute Arbitrary Queries

==
21. SECRETS MANAGEMENT
==

Secrets

• JWT Secret
• Database Passwords
• OAuth Keys
• API Keys

Storage

HashiCorp Vault

Alternative

AWS Secrets Manager

Never Store In

✗ Source Code
✗ GitHub
✗ Environment Files in Repository

==
22. MONITORING & INCIDENT RESPONSE
==

Monitoring Stack

• Prometheus
• Grafana
• Loki
• Alertmanager

Alerts

• Failed Logins
• Traffic Spikes
• Fraud Detection
• API Abuse

==
======
23. COMPLIANCE REQUIREMENTS
==
======

Recommended Standards

• ISO 27001
• OWASP ASVS
• OWASP Top 10
• SOC 2 (Future)
• Government Security Guidelines

==
======
24. DISASTER RECOVERY SECURITY
==
======

Backups

Database:
Every 6 Hours

Media:
Daily

Audit Logs:
Every 6 Hours

Recovery Objectives

RPO

15 Minutes

RTO

1 Hour

Recovery Validation

Quarterly DR Testing

==
END OF DOCUMENT
==

==
FRONTEND SPECIFICATION DOCUMENT
Civic Intelligence & Municipal Operations Platform (CIMOP)
Version: 1.0
Document Type: Frontend Engineering Specification
==

TABLE OF CONTENTS

==
1. FRONTEND VISION
==

The frontend must provide a modern, government-grade digital experience that is:

• Fast
• Responsive
• Mobile-first
• Accessible
• Intuitive
• Scalable

Primary Goals

1. Simplify complaint submission.
2. Reduce citizen effort.
3. Improve transparency.
4. Enable efficient employee operations.
5. Deliver real-time visibility.
6. Support large-scale municipal deployments.

==
======
2. FRONTEND TECHNOLOGY STACK
==
======

Framework

Next.js 15

Reason

• SSR
• SEO
• Fast Routing
• Production Ready

Language

TypeScript

Reason

• Type Safety
• Better Maintainability

Styling

Tailwind CSS

Reason

• Rapid Development
• Consistent Design

UI Components

shadcn/ui

Reason

• Accessible
• Modern

• Production Grade

Forms

React Hook Form

Zod

Maps

Mapbox GL JS

Reason

• Advanced GIS Features
• Heatmaps
• Clustering

Charts

Recharts

Reason

• Dashboard Friendly

Real-Time

Socket.IO Client

==
3. FRONTEND ARCHITECTURE
==

src/

```
├── app/
├── components/
├── features/
├── hooks/
├── services/
├── store/
├── lib/
├── types/
├── utils/
├── styles/
```

==
======
4. DESIGN SYSTEM
==
======

COLOR PALETTE

Primary

Blue 600

Purpose

Government Trust

Success

Green 600

Warning

Orange 500

Danger

Red 600

Neutral

Gray Scale

==
======
TYPOGRAPHY
==
======

Headings

Inter

Font Weight

600-700

Body

Inter

Font Weight

400–500

==
======
SPACING SYSTEM
==
======

4px Scale

4
8
12
16
24
32
48
64

==
======
5. USER EXPERIENCE PRINCIPLES
==
======

PRINCIPLE 1

Minimum clicks to submit complaint.

PRINCIPLE 2

Complaint status always visible.

PRINCIPLE 3

Real-time updates.

PRINCIPLE 4

Mobile-first design.

PRINCIPLE 5

Accessibility-first.

==
======
6. ROUTING ARCHITECTURE
==
======

Public Routes

/

/about

/contact

/help

/faq

/login

/register

==
======
Citizen Routes
==
======

/dashboard

/complaints

/complaints/create

/complaints/history

/complaints/[id]

/profile

/settings

/notifications

==
======
Employee Routes
==
======

/employee/dashboard

/employee/tasks

/employee/tasks/[id]

/employee/history

/profile

==
======
Manager Routes
==
======

/manager/dashboard

/manager/complaints

/manager/employees

/manager/reports

==
======
Administrator Routes
==
======

/admin/dashboard

/admin/analytics

/admin/departments

/admin/zones

/admin/users

==
======
Super Admin Routes
==
======

/super-admin/dashboard

/security

/roles

/audit-logs

/system-config

==
======
7. STATE MANAGEMENT ARCHITECTURE
==
======

State Manager

Zustand

Reason

• Lightweight
• Easy Scaling

Stores

Auth Store

Complaint Store

Notification Store

Map Store

Analytics Store

User Store

==
8. CITIZEN PORTAL
==

MAIN GOAL

Enable citizens to quickly report and track issues.

==
PAGE 1
Citizen Dashboard
==

Widgets

• Total Complaints
• Active Complaints
• Resolved Complaints
• Escalated Complaints

Recent Activity

Complaint Timeline

Map Widget

Nearby Complaints

==
PAGE 2
Create Complaint

==
======

Step 1

Select Category

Step 2

Describe Issue

Fields

• Title
• Description

Step 3

Upload Evidence

Images

Videos

Voice Recording

Step 4

Location

GPS Detection

OR

Map Selection

Step 5

Review & Submit

==
======
PAGE 3
Complaint Details
==
======

Sections

Complaint Information

AI Analysis Result

Status Timeline

Department Information

Assigned Worker

Evidence Gallery

Resolution Proof

Citizen Feedback

==
======
PAGE 4
Complaint History
==
======

Filters

Status

Category

Date

Priority

Search

==
======
PAGE 5
Notifications
==
======

Notification Types

Status Updates

Escalations

Assignments

Announcements

==
======
9. EMPLOYEE PORTAL
==
======

MAIN GOAL

Resolve assigned issues efficiently.

==
======
Employee Dashboard
==
======

Widgets

Assigned Today

Pending Tasks

Completed Tasks

Average Resolution Time

==
======
Task List
==
======

Filters

Priority

Area

Status

==
======
Task Detail Page
==
======

Sections

Complaint Information

Location

Navigation

Citizen Notes

Evidence

AI Analysis

==
======
Actions
==
======

Accept Task

Update Status

Upload Work Evidence

Mark Resolved

==
======
10. DEPARTMENT MANAGER PORTAL
==
======

Dashboard

Complaint Queue

Worker Allocation

SLA Monitoring

Department Analytics

==
======
Complaint Management
==
======

Actions

Assign Worker

Reassign Worker

Escalate Complaint

Approve Closure

==
======
11. CITY ADMINISTRATOR PORTAL
==
======

Dashboard Purpose

City-wide visibility.

==
======
Widgets
==
======

Infrastructure Health Index

Complaint Volume

Resolution Rate

Department Performance

Citizen Satisfaction

Hotspot Heatmap

==
Analytics Pages
==

Complaint Trends

Resolution Trends

Zone Analytics

Department Analytics

Infrastructure Analytics

==
12. SUPER ADMIN PORTAL
==

Features

User Management

Role Management

Permission Management

AI Configuration

System Settings

Security Dashboard

Audit Logs

==
13. SHARED COMPONENTS
==

Core Components

Button

Input

Textarea

Select

Dialog

Drawer

Sheet

Card

Badge

Tabs

Tooltip

Table

Pagination

==
Business Components
==

Complaint Card

Complaint Timeline

Status Badge

Department Badge

Priority Badge

Notification Card

Analytics Widget

Map Widget

==
14. MAPS & GIS UI
==

Map Features

Current Location

Complaint Marker

Heatmap Layer

Cluster Layer

Department Zones

Infrastructure Zones

==
======
Citizen Features
==
======

Choose Location

View Nearby Issues

Track Complaint Location

==
======
Admin Features
==
======

Hotspots

Coverage Zones

Issue Density Maps

==
======
15. REAL-TIME FEATURES
==
======

Socket.IO Events

Complaint Created

Complaint Assigned

Complaint Updated

Complaint Resolved

Notification Created

==
======
Real-Time UI Updates
==
======

Status Changes

Notification Panel

Dashboard Metrics

Assignment Updates

==
16. NOTIFICATION UI
==

Channels

In-App

Push

Email

SMS

==
Notification Center
==

Unread Count

Read History

Priority Labels

Action Buttons

==
17. AI ASSISTANT UI
==

IMPORTANT

This is NOT a general chatbot.

==
Allowed Topics
==

Complaint Creation

Complaint Tracking

Municipal Services

Department Information

Resolution Status

==
======
Blocked Topics
==
======

Programming

Math

Education

General Knowledge

Career Advice

Entertainment

==
======
Chat Interface
==
======

Simple Chat Panel

Quick Actions

Create Complaint

Track Complaint

View Department

==
======
18. ACCESSIBILITY STANDARDS
==
======

Compliance

WCAG 2.1 AA

==
======
Features
==
======

Keyboard Navigation

Screen Reader Support

Focus Indicators

Color Contrast Validation

Responsive Font Sizes

==
======
19. MOBILE RESPONSIVENESS
==
======

Breakpoints

Mobile

Tablet

Laptop

Desktop

Ultra Wide

==
======
Mobile Priority Pages
==
======

Complaint Submission

Complaint Tracking

Notifications

Maps

==
======
20. PERFORMANCE REQUIREMENTS
==
======

Initial Load

< 2 Seconds

Dashboard Load

< 2 Seconds

Complaint Submission

< 1 Second UI Response

Lighthouse Targets

Performance > 90

Accessibility > 95

Best Practices > 95

SEO > 90

==
======
FRONTEND DEVELOPMENT PHASES
==
======

PHASE 1

Authentication

Citizen Portal

Complaint Submission

Complaint Tracking

PHASE 2

Employee Portal

Manager Portal

Notifications

PHASE 3

Admin Dashboard

GIS Features

Analytics

PHASE 4

AI Assistant

Predictive Analytics

Advanced Reporting

==
END OF DOCUMENT
==

==
BACKEND SPECIFICATION DOCUMENT
Civic Intelligence & Municipal Operations Platform (CIMOP)
Version: 1.0
Document Type: Backend Engineering Specification
Architecture: Modular Monolith (Phase 1) → Microservices (Phase 2)
==

TABLE OF CONTENTS

==
1. BACKEND VISION
==

The backend acts as the operational brain of the platform.

Objectives:

• Process 100,000 complaints/day
• Handle AI analysis workloads
• Support real-time updates
• Maintain auditability
• Provide secure APIs
• Enable future smart-city integrations

Core Design Goals:

✓ Maintainable
✓ Scalable
✓ Testable
✓ Secure
✓ Event-Driven
✓ AI Ready

==
2. TECHNOLOGY STACK
==

Primary Framework

FastAPI

Reason

• High Performance
• Async Support
• OpenAPI Generation
• Python AI Ecosystem

Programming Language

Python 3.13+

ORM

SQLAlchemy 2.x

Migration Tool

Alembic

Validation

Pydantic v2

Database

PostgreSQL + PostGIS

Cache

Redis

Message Broker

RabbitMQ

Task Queue

Celery

Vector Search

Qdrant

Object Storage

MinIO (Local)
AWS S3 (Production)

==
3. ARCHITECTURAL APPROACH
==

PHASE 1

Modular Monolith

Reason:

• Faster development
• Easier debugging
• Lower infrastructure complexity

Structure:

Single FastAPI application

Multiple internal modules

Shared database

Shared Redis

Shared RabbitMQ

PHASE 2

Microservices

Split Services:

- Auth Service
- Complaint Service
- AI Service
- Analytics Service
- Notification Service

==
4. BACKEND LAYER ARCHITECTURE
==

Request Flow

```
Client
 |
 v
API Router
 |
 v
Controller Layer
 |
 v
Service Layer
 |
 v
Repository Layer
 |
 v
Database
```

Layer Responsibilities

Controller Layer

- Request validation
- Authentication

• Response formatting

Service Layer

• Business logic
• Workflow orchestration
• AI integration

Repository Layer

• Database interaction
• Query optimization

Infrastructure Layer

• Redis
• RabbitMQ
• File Storage
• External APIs

==
5. FASTAPI PROJECT STRUCTURE
==

```
backend/

├── app/
│
├── api/
│   ├── citizen/
│   ├── employee/
│   ├── manager/
│   ├── admin/
│   └── auth/
│
├── core/
│   ├── config/
│   ├── security/
│   ├── middleware/
│   ├── exceptions/
│   └── logging/
│
├── modules/
│
│   ├── users/
│   ├── complaints/
│   ├── departments/
│   ├── assignments/
│   ├── notifications/
│   ├── analytics/
│   │
```

```
│   ├── ai/
│   ├── gis/
│   ├── fraud/
│   └── audit/
│
├── infrastructure/
│
│   ├── redis/
│   ├── rabbitmq/
│   ├── storage/
│   ├── email/
│   ├── sms/
│   └── websocket/
│
├── workers/
│
├── tests/
│
└── main.py
```

==
======
6. CORE SERVICES
==
======

SERVICE 1

Authentication Service

Responsibilities

• Login
• Registration
• Token Generation
• Password Reset
• OAuth

SERVICE 2

User Service

Responsibilities

• Citizen Management
• Employee Management
• Profile Updates

SERVICE 3

Complaint Service

Responsibilities

• Complaint Creation
• Complaint Updates
• Complaint Search
• Complaint Tracking

SERVICE 4

Assignment Service

Responsibilities

• Department Assignment
• Worker Assignment
• Reassignment

SERVICE 5

Notification Service

Responsibilities

• Email
• SMS
• Push
• In-App

SERVICE 6

Analytics Service

Responsibilities

• KPI Generation
• Trend Analysis

SERVICE 7

Audit Service

Responsibilities

• Security Logs
• Action Logs

==
======
7. DOMAIN MODULES
==
======

USER MODULE

Entities

User
Role
Permission

COMPLAINT MODULE

Entities

Complaint
ComplaintStatus
ComplaintMedia
ComplaintLocation

DEPARTMENT MODULE

Entities

Department
DepartmentZone

ASSIGNMENT MODULE

Entities

Assignment
WorkerTask

AI MODULE

Entities

AIAnalysis
Classification
FraudScore

==
======
8. API GATEWAY DESIGN
==
======

Responsibilities

• Authentication
• Rate Limiting
• Logging
• Routing

Nginx

Acts As

Reverse Proxy

Flow

```
Client
 |
 v
Nginx
 |
 v
FastAPI
```

==
======
9. BUSINESS WORKFLOW ENGINE
==
======

Complaint Workflow

```
Complaint Created
 |
 AI Analysis
 |
 Department Assignment
 |
 Worker Assignment
 |
 In Progress
 |
 Resolution Upload
 |
 Verification
 |
 Closed
```

Escalation Workflow

```
Complaint
 |
 SLA Breach
 |
 Escalation Trigger
 |
 Manager Review
 |
 Reassignment
```

==

10. EVENT-DRIVEN ARCHITECTURE

==

Events

ComplaintCreated

ComplaintUpdated

ComplaintAssigned

ComplaintResolved

ComplaintEscalated

FraudDetected

NotificationCreated

Event Flow

```
Complaint Created
 |
 RabbitMQ Event
 |
 AI Worker Consumes
 |
 Analysis Complete
 |
 Assignment Event
 |
 Notification Event
```

==

11. BACKGROUND PROCESSING

==
======

Celery Workers

Worker Type 1

AI Worker

Tasks

• Classification
• Duplicate Detection
• Fraud Analysis

Worker Type 2

Notification Worker

Tasks

• Email
• SMS
• Push

Worker Type 3

Analytics Worker

Tasks

• KPI Updates
• Report Generation

Worker Type 4

Media Worker

Tasks

• Compression
• Thumbnail Generation
• Virus Scan

==
======
12. REDIS ARCHITECTURE
==
======

Usage

Cache Layer

Store

Frequently Accessed Complaints

Dashboard Data

Sessions

JWT Blacklist

Notification Cache

Expiration Strategy

Dashboard Cache

5 Minutes

Complaint Cache

10 Minutes

Session Cache

30 Days

==
13. RABBITMQ ARCHITECTURE
==

Queues

complaint.created

complaint.analyzed

complaint.assigned

complaint.resolved

notification.queue

analytics.queue

fraud.queue

Dead Letter Queue

dlq.complaints

dlq.notifications

dlq.analytics

==
14. SEARCH ARCHITECTURE
==

Phase 1

PostgreSQL Full Text Search

Phase 2

OpenSearch

Features

• Advanced Filtering
• Full-Text Search
• Aggregations

==
15. FILE MANAGEMENT ARCHITECTURE
==

Supported Files

Images

Videos

Audio

Storage Flow

```
Upload
 |
 Virus Scan
 |
```

```
 Validation
 |
 Compression
 |
 MinIO/S3
 |
 Store Metadata
```


Image Processing

Resize

Generate Thumbnails

Extract Metadata

==
======
16. NOTIFICATION ARCHITECTURE
==
======

Channels

In-App

Push

SMS

Email

Notification Flow

```
Event
 |
 RabbitMQ
 |
 Notification Worker
 |
 Delivery
```

==
======
17. GIS PROCESSING ARCHITECTURE
==
======

GIS Database

PostGIS

Functions

Spatial Queries

Distance Calculation

Heatmaps

Zone Mapping

Cluster Detection

Example

Find nearest department

Find nearby complaints

Generate hotspot areas

==

18. ANALYTICS PROCESSING

==

Real-Time Metrics

Complaint Count

Resolution Count

Open Complaints

Batch Metrics

Daily Reports

Monthly Reports

Department Performance

Generated KPIs

Average Resolution Time

Citizen Satisfaction

SLA Compliance

==
======
19. AUDIT ARCHITECTURE
==
======

Audit Events

Login

Logout

Complaint Creation

Status Change

Assignment

Role Change

Permission Change

Stored Fields

User ID

Timestamp

IP Address

Action

Entity Type

Entity ID

==
======
20. PERFORMANCE STRATEGY
==
======

Target Metrics

API Response

<300ms

Complaint Submission

<2 Seconds

AI Classification

<5 Seconds

Dashboard Load

<2 Seconds

==
======
21. SCALABILITY STRATEGY
==
======

Horizontal Scaling

API Servers

Celery Workers

RabbitMQ Nodes

Redis Nodes

Database Scaling

Read Replicas

Partitioning

Connection Pooling

Complaint Partitioning

By

Year

Month

Tenant

==
======
22. MIGRATION PATH TO MICROSERVICES
==
======

PHASE 1

Modular Monolith

PHASE 2

Extract

AI Service

Notification Service

Analytics Service

PHASE 3

Extract

Auth Service

Complaint Service

PHASE 4

Service Mesh

Kubernetes

Istio

Final Architecture

Independent Services

Independent Databases

Independent Scaling

==
======

BACKEND DEVELOPMENT PRIORITY
==
======

Sprint 1

Authentication

Users

Roles

Sprint 2

Complaint Module

Media Upload

Tracking

Sprint 3

Assignment System

Notifications

Sprint 4

AI Integration

Sprint 5

Analytics

GIS

Sprint 6

Fraud Detection

Predictive Maintenance

==
======
END OF DOCUMENT
==
======

MACHINE LEARNING SPECIFICATION DOCUMENT

AI-Powered Civic Intelligence & Municipal Operations Platform

Version: 1.0

Document Type: Machine Learning System Design

Author: ML Architecture Team

---

1. MACHINE LEARNING VISION

The ML system serves as the intelligence layer of the Civic Intelligence Platform.

Objectives:

1. Automatically understand complaints.
2. Categorize complaints.
3. Detect infrastructure issues from images.
4. Detect duplicate complaints.
5. Prioritize complaints.
6. Detect fraud and abuse.
7. Route complaints automatically.
8. Predict future infrastructure failures.
9. Generate city-level intelligence.
10. Assist municipal departments with decision making.

The ML system is NOT designed as a general AI assistant.

The entire ML stack operates only within civic and municipal domains.

---

2. MACHINE LEARNING SUBSYSTEMS

The platform consists of 8 major ML engines.

ML-01 Complaint Classification Engine

ML-02 Computer Vision Detection Engine

ML-03 Duplicate Complaint Detection Engine

ML-04 Complaint Priority Engine

ML-05 Fraud Detection Engine

ML-06 Complaint Routing Engine

ML-07 Predictive Infrastructure Engine

ML-08 Civic AI Domain Restriction Engine

---

3. END-TO-END AI PIPELINE

Citizen Input

(Text + Image + Voice + GPS)

↓

Input Validation

↓

Voice Processing

↓

Image Analysis

↓

Text Analysis

↓

Complaint Classification

↓

Severity Estimation

↓

Duplicate Detection

↓

Fraud Detection

↓

Department Prediction

↓

Priority Scoring

↓

Master AI Decision Engine

↓

Complaint Creation

↓

Task Assignment

↓

Dashboard Analytics

---

4. COMPLAINT CLASSIFICATION ENGINE

Purpose:

Identify complaint category automatically.

Supported Categories:

Road Issues

Garbage Issues

Drainage Issues

Water Supply Issues

Street Light Issues

Traffic Signal Issues

Public Toilet Issues

Park Maintenance

Public Property Damage

Electrical Infrastructure

Other

---

5. NLP MODEL ARCHITECTURE

Input:

Complaint text

Examples:

"There is a huge pothole near city mall"

"Streetlight not working from last 5 days"

"Garbage pile causing smell"

Output:

Category

Confidence Score

Department

Severity Signal

---

6. MODEL COMPARISON

OPTION 1

DistilBERT

Pros:

Fast

Small model

Low memory

Easy deployment

Cons:

Lower accuracy

Expected Accuracy:

89-92%

OPTION 2

BERT Base

Pros:

High accuracy

Strong language understanding

Cons:

Heavy inference cost

Expected Accuracy:

93-95%

OPTION 3

RoBERTa

Pros:

Best contextual understanding

High classification performance

Cons:

More compute intensive

Expected Accuracy:

95-97%

OPTION 4

MiniLM

Pros:

Very fast

Low latency

Excellent embeddings

Cons:

Slightly lower classification power

Expected Accuracy:

91-94%

RECOMMENDED MODEL

Production Classification Model:

RoBERTa Base

Reason:

Highest F1 Score

Better handling of citizen-written complaints

Strong multilingual fine tuning

Best category prediction accuracy

---

7. DATASET REQUIREMENTS

Minimum Training Size

100,000 Complaints

Ideal Size

500,000 Complaints

Enterprise Scale

2M+ Complaints

Required Fields

Complaint Text

Category

Department

Location

Resolution Status

Priority

Language

Timestamp

---

8. LABELING STRATEGY

Manual Municipal Annotation

Labelers:

Municipal officers

Domain experts

QA team

Annotation Guidelines:

Road -> potholes, cracks, damaged roads

Garbage -> waste accumulation

Drainage -> sewage overflow

Water -> leakage, pipeline issues

Electricity -> exposed wires, transformer issues

Traffic -> signal malfunction

Etc.

---

9. TRAINING PIPELINE

Raw Data

↓

Cleaning

↓

Language Detection

↓

Tokenization

↓

Label Verification

↓

Train/Test Split

↓

Model Training

↓

Evaluation

↓

Model Registry

↓

Deployment

---

10. EVALUATION METRICS

Precision

Target:

95%+

Recall

Target:

95%+

F1 Score

Target:

95%+

Classification Accuracy

Target:

95%+

Inference Time

Target:

< 200ms

---

11. COMPUTER VISION ENGINE

Purpose:

Detect infrastructure issues directly from images.

Supported Detection Classes

Pothole

Garbage

Water Leakage

Broken Street Light

Damaged Public Property

Drainage Overflow

Broken Bench

Open Manhole

Illegal Dumping

Road Crack

---

12. OBJECT DETECTION MODEL COMPARISON

YOLOv8

Accuracy:

Excellent

Speed:

Excellent

Production Ready:

Yes

YOLOv11

Accuracy:

Best

Speed:

Excellent

Production Ready:

Yes

EfficientNet

Object Detection:

No

Classification Only

Faster R-CNN

Accuracy:

High

Speed:

Slow

Not ideal for real-time systems

RECOMMENDED MODEL

YOLOv11

Reason:

Highest detection accuracy

Fast inference

Real-time deployment ready

Best infrastructure defect detection

---

13. IMAGE DATASET REQUIREMENTS

Minimum:

50,000 Images

Target:

500,000 Images

Enterprise:

2M+ Images

---

14. IMAGE ANNOTATION

Tools:

CVAT

Label Studio

Roboflow

Annotation Type:

Bounding Boxes

Polygon Segmentation

Severity Labels

---

15. DATA AUGMENTATION

Rotation

Cropping

Brightness Changes

Weather Simulation

Noise Injection

Perspective Transform

Blur Simulation

Night-time Simulation

---

16. VOICE COMPLAINT ENGINE

Pipeline

Voice Input

↓

Speech To Text

↓

Language Detection

↓

Translation (Optional)

↓

Complaint Classification

↓

Complaint Creation

---

17. RECOMMENDED SPEECH MODELS

Whisper Large v3

Best Accuracy

Multilingual

Offline Deployable

Recommended

Alternative:

Google Speech API

Azure Speech

AWS Transcribe

---

18. DUPLICATE COMPLAINT DETECTION

Business Goal

50 citizens report same pothole

System should create:

1 Master Complaint

50 Citizen References

---

19. DUPLICATE DETECTION PIPELINE

Complaint Text

↓

Embedding Generation

↓

Location Matching

↓

Image Similarity

↓

Vector Search

↓

Similarity Score

↓

Duplicate Decision

---

20. EMBEDDING MODEL

Recommended:

Sentence Transformers

all-MiniLM-L6-v2

Advantages:

Fast

High semantic quality

Lightweight

---

21. VECTOR DATABASE COMPARISON

Pinecone

Managed

Expensive

Qdrant

Fast

Open Source

Excellent Performance

---

pgvector

Built inside PostgreSQL

Simple deployment

Lower complexity

RECOMMENDED

Qdrant

Reason:

Scalable

High performance

Open source

Excellent similarity search

---

22. DUPLICATE SCORING FORMULA

Text Similarity

40%

Location Similarity

35%

Image Similarity

25%

Duplicate Threshold:

0.85+

Likely Duplicate

0.70-0.85

Human Review

Below 0.70

New Complaint

---

23. PRIORITY PREDICTION ENGINE

Priority Classes

Low

Medium

High

Critical

---

24. PRIORITY SCORING FEATURES

Issue Type

Severity

Population Impact

Road Importance

School Nearby

Hospital Nearby

Electrical Risk

Flood Risk

Duplicate Count

Historical Escalations

---

25. PRIORITY FORMULA

Priority Score

=
Infrastructure Risk (35%)

*

Citizen Impact (25%)

*

Safety Risk (25%)

*

Complaint Density (15%)

---

26. FRAUD DETECTION ENGINE

Detect:

Spam Complaints

Fake Complaints

Bot Accounts

Mass Abuse

Location Spoofing

Duplicate Accounts

Image Manipulation

---

27. FRAUD FEATURES

Complaints Per Hour

Account Age

Resolution History

Device Fingerprint

IP Reputation

GPS Consistency

Image Metadata

Complaint Similarity

---

28. FRAUD MODELS

Isolation Forest

XGBoost

Random Forest

Autoencoders

RECOMMENDED

XGBoost + Isolation Forest

---

29. ROUTING ENGINE

Purpose:

Automatically assign department.

Example:

Pothole

→ Road Department

Water Leakage

→ Water Department

Street Light

→ Electrical Department

---

30. PREDICTIVE MAINTENANCE ENGINE

Future AI Capability

Predict:

Future Potholes

Future Garbage Hotspots

Future Drainage Failures

Future Water Leakages

Future Streetlight Failures

---

31. PREDICTIVE FEATURES

Historical Complaints

Rainfall

Traffic Density

Infrastructure Age

Maintenance Records

Weather Conditions

Population Density

Road Usage

---

32. PREDICTIVE MODELS

XGBoost

LightGBM

CatBoost

Random Forest

LSTM

RECOMMENDED

LightGBM

Reason:

Fast

Accurate

Handles tabular data perfectly

---

33. DOMAIN RESTRICTION AI

Layer 1

Intent Classification

Layer 2

Domain Validation

Layer 3

Prompt Filtering

Layer 4

Output Validation

Layer 5

Moderation Layer

---

34. ALLOWED TOPICS

Complaint Creation

Complaint Tracking

Municipal Services

Departments

Infrastructure Issues

Issue Resolution

---

35. BLOCKED TOPICS

Programming

Coding

Homework

Mathematics

Story Writing

Translation

Career Advice

Entertainment

General Knowledge

---

36. MLOPS ARCHITECTURE

Data Collection

↓

Data Lake

↓

Training Pipeline

↓

Model Registry

↓

Model Validation

↓

Deployment

↓

Monitoring

↓

Retraining

---

37. MODEL REGISTRY

MLflow

Version Control

Model Tracking

Experiment Tracking

Deployment Tracking

---

38. MODEL MONITORING

Data Drift

Concept Drift

Prediction Accuracy

Latency

Failure Rate

Duplicate Detection Accuracy

Fraud Detection Accuracy

---

39. RETRAINING STRATEGY

Complaint Classifier

Monthly

Vision Model

Monthly

Fraud Model

Weekly

Priority Model

Bi-Weekly

Predictive Models

Monthly

---

40. TARGET PRODUCTION METRICS

Complaint Classification Accuracy

95%+

Image Detection mAP

90%+

Duplicate Detection Accuracy

95%+

Fraud Detection Precision

92%+

Priority Prediction Accuracy

90%+

Average AI Inference

< 500ms

Platform Scale

100,000 Complaints Daily

1 Million Citizens

99.9% Availability

END OF MACHINE LEARNING SPECIFICATION DOCUMENT

FEATURE TICKET LIST DOCUMENT

AI-Powered Civic Intelligence & Municipal Operations Platform

Version: 1.0

Document Type: Product Backlog & Engineering Ticket Breakdown

Purpose:
This document serves as the master engineering backlog for product, frontend, backend, AI/ML, DevOps, QA, and security teams.

TICKET STRUCTURE

Epic

→ Feature
→ User Story
→ Engineering Tasks

Priority Levels:

P0 = Critical

P1 = High

P2 = Medium

P3 = Low

EPIC 1
USER AUTHENTICATION & ACCESS MANAGEMENT

Priority: P0

Goal:
Provide secure authentication, authorization, and account management.

FEATURE 1.1
Citizen Registration

User Story:

As a citizen,
I want to create an account,
So that I can report and track complaints.

Tasks:

AUTH-001
Create registration API

AUTH-002
Validate email

AUTH-003
Validate phone number

AUTH-004
Password hashing

AUTH-005
OTP verification

AUTH-006
Citizen profile creation

AUTH-007

Welcome notification

AUTH-008
Account activation flow

FEATURE 1.2
Login System

AUTH-009
Email login

AUTH-010
Phone login

AUTH-011
JWT access token

AUTH-012
Refresh token

AUTH-013
Session management

AUTH-014
Logout functionality

AUTH-015
Multi-device support

FEATURE 1.3
Password Recovery

AUTH-016
Forgot password API

AUTH-017
OTP verification

AUTH-018
Reset password flow

AUTH-019
Password policy validation

FEATURE 1.4
Role Based Access Control

AUTH-020
Citizen permissions

AUTH-021
Field Worker permissions

AUTH-022
Department Manager permissions

AUTH-023
City Administrator permissions

AUTH-024
Super Admin permissions

AUTH-025
Permission middleware

AUTH-026
Role assignment engine

EPIC 2
CITIZEN PROFILE MANAGEMENT

Priority: P1

PROFILE-001
View profile

PROFILE-002
Update profile

PROFILE-003
Profile picture upload

PROFILE-004
Address management

PROFILE-005
Notification preferences

PROFILE-006
Language preferences

PROFILE-007
Complaint statistics

PROFILE-008
Account deactivation

EPIC 3
COMPLAINT SUBMISSION SYSTEM

Priority: P0

Goal:
Allow citizens to submit complaints using multiple input methods.

FEATURE 3.1
Text Complaint Submission

COMP-001
Complaint form UI

COMP-002
Description validation

COMP-003
Category suggestion

COMP-004
Draft saving

COMP-005
Complaint creation API

FEATURE 3.2
Image Upload Complaints

COMP-006
Single image upload

COMP-007
Multiple image upload

COMP-008
Image compression

COMP-009
Virus scanning

COMP-010
Image storage integration

FEATURE 3.3
Video Upload Complaints

COMP-011
Video upload

COMP-012
Video compression

COMP-013
Video streaming support

COMP-014
Video moderation checks

FEATURE 3.4
GPS Complaint Submission

COMP-015
GPS capture

COMP-016
Location validation

COMP-017
Reverse geocoding

COMP-018
Map pin placement

FEATURE 3.5
Voice Complaint Submission

COMP-019
Voice recording

COMP-020
Speech-to-text conversion

COMP-021
Transcript validation

COMP-022
Voice complaint creation

EPIC 4
AI COMPLAINT ANALYZER

Priority: P0

Goal:
Automatically understand complaints.

FEATURE 4.1
Complaint Classification Engine

AI-001
Text preprocessing pipeline

AI-002
Classification model service

AI-003
Confidence scoring

AI-004
Category prediction

AI-005
Model monitoring

FEATURE 4.2
Department Prediction

AI-006
Department mapping rules

AI-007
Department prediction API

AI-008
Fallback routing logic

FEATURE 4.3
Severity Analysis

AI-009
Severity scoring model

AI-010
Severity calibration

AI-011
Critical issue detection

FEATURE 4.4
AI Decision Engine

AI-012
Decision orchestration service

AI-013
Inference workflow

AI-014
Decision logging

EPIC 5
COMPUTER VISION ENGINE

Priority: P0

VISION-001
Image preprocessing

VISION-002
YOLO inference service

VISION-003
Pothole detection

VISION-004
Garbage detection

VISION-005
Drainage issue detection

VISION-006
Water leakage detection

VISION-007
Street light detection

VISION-008
Open manhole detection

VISION-009
Damage severity estimation

VISION-010
Bounding box visualization

VISION-011
Detection confidence scoring

EPIC 6
DUPLICATE COMPLAINT DETECTION

Priority: P0

DUP-001
Embedding generation

DUP-002
Vector search integration

DUP-003
Location similarity engine

DUP-004
Image similarity engine

DUP-005
Duplicate scoring

DUP-006
Master complaint generation

DUP-007
Complaint linking

DUP-008
Duplicate review dashboard

EPIC 7
FRAUD DETECTION SYSTEM

Priority: P0

FRAUD-001
Spam detection

FRAUD-002
Bot detection

FRAUD-003
Device fingerprinting

FRAUD-004
IP reputation checks

FRAUD-005
Risk scoring engine

FRAUD-006
Fraud alert generation

FRAUD-007
Suspicious account review

EPIC 8
COMPLAINT WORKFLOW MANAGEMENT

Priority: P0

WORK-001
Complaint lifecycle engine

WORK-002
Status transition rules

WORK-003
Assignment engine

WORK-004
Complaint ownership

WORK-005
Escalation workflows

WORK-006
Closure workflows

WORK-007
Reopen complaint flow

EPIC 9
FIELD WORKER OPERATIONS

Priority: P0

FIELD-001
Worker dashboard

FIELD-002
Assigned tasks list

FIELD-003
Task acceptance

FIELD-004
Task progress update

FIELD-005
Resolution image upload

FIELD-006
Resolution notes

FIELD-007
Work completion submission

FIELD-008
Mobile optimized UI

EPIC 10
DEPARTMENT MANAGEMENT

Priority: P1

DEPT-001
Department dashboard

DEPT-002
Department workload view

DEPT-003
Team assignment

DEPT-004
Performance analytics

DEPT-005
Escalation handling

DEPT-006
Department reporting

EPIC 11
REAL-TIME NOTIFICATION SYSTEM

Priority: P1

NOTIF-001
Push notifications

NOTIF-002
Email notifications

NOTIF-003
SMS notifications

NOTIF-004
In-app notifications

NOTIF-005
Notification templates

NOTIF-006
Delivery tracking

EPIC 12
COMPLAINT TRACKING SYSTEM

Priority: P0

TRACK-001
Live complaint status

TRACK-002
Status timeline

TRACK-003
Resolution timeline

TRACK-004
Tracking API

TRACK-005
Citizen tracking dashboard

EPIC 13
MAPS & GIS PLATFORM

Priority: P1

GIS-001
Map integration

GIS-002
Complaint markers

GIS-003
Cluster visualization

GIS-004
Heatmap generation

GIS-005
Department zones

GIS-006
Infrastructure hotspots

GIS-007
Spatial analytics

EPIC 14
ADMIN ANALYTICS DASHBOARD

Priority: P1

ADMIN-001
City overview dashboard

ADMIN-002
Complaint analytics

ADMIN-003
Department analytics

ADMIN-004
Resolution analytics

ADMIN-005
Infrastructure health score

ADMIN-006
Trend analysis

ADMIN-007
Custom reports

ADMIN-008
Dashboard exports

EPIC 15
PREDICTIVE MAINTENANCE AI

Priority: P2

PRED-001
Historical data pipeline

PRED-002
Feature engineering

PRED-003
Prediction model

PRED-004
Hotspot prediction

PRED-005
Failure forecasting

PRED-006
Maintenance recommendations

EPIC 16
DOMAIN RESTRICTED CIVIC AI ASSISTANT

Priority: P0

CIVICAI-001
Intent classification

CIVICAI-002
Domain validation

CIVICAI-003
Prompt guardrails

CIVICAI-004
Output filtering

CIVICAI-005
Moderation system

CIVICAI-006
Scope restriction testing

CIVICAI-007
Prompt injection protection

EPIC 17
REPORTING & EXPORTS

Priority: P2

REPORT-001
PDF exports

REPORT-002
Excel exports

REPORT-003
Department reports

REPORT-004
Citizen reports

REPORT-005
Analytics exports

EPIC 18
AUDIT & COMPLIANCE

Priority: P1

AUDIT-001
Audit log service

AUDIT-002
Admin activity tracking

AUDIT-003
Complaint history logs

AUDIT-004
Data retention policies

AUDIT-005
Compliance reports

EPIC 19

SECURITY PLATFORM

Priority: P0

SEC-001
JWT security

SEC-002
RBAC middleware

SEC-003
Rate limiting

SEC-004
API security

SEC-005
Encryption at rest

SEC-006
Encryption in transit

SEC-007
File upload security

SEC-008
WAF integration

SEC-009
Security monitoring

EPIC 20
DEVOPS & INFRASTRUCTURE

Priority: P0

DEVOPS-001
Docker setup

DEVOPS-002
Kubernetes deployment

DEVOPS-003
CI/CD pipeline

DEVOPS-004
Redis setup

DEVOPS-005
RabbitMQ setup

DEVOPS-006
Monitoring stack

DEVOPS-007
Logging pipeline

DEVOPS-008
Disaster recovery

DEVOPS-009
Auto scaling

DEVOPS-010
Backup strategy

EPIC 21
MOBILE APPLICATION

Priority: P2

MOBILE-001
Citizen mobile app

MOBILE-002
Worker mobile app

MOBILE-003
Offline mode

MOBILE-004
Push notifications

MOBILE-005
GPS tracking

EPIC 22
QA & TESTING

Priority: P0

QA-001
Unit testing

QA-002
Integration testing

QA-003
Load testing

QA-004
Security testing

QA-005
AI model validation

QA-006
Regression testing

QA-007
UAT testing

ENGINEERING ESTIMATION

Total Epics:
22

Estimated Features:
120+

Estimated Engineering Tickets:
300+

Estimated Development Timeline:
8–12 Months

Recommended Team Size:
15–25 Engineers

Backend Engineers: 4

Frontend Engineers: 3

AI/ML Engineers: 3

DevOps Engineers: 2

QA Engineers: 3

Product Manager: 1

UI/UX Designer: 1

Technical Architect: 1

END OF FEATURE TICKET LIST DOCUMENT

DATABASE DESIGN DOCUMENT

AI-Powered Civic Intelligence & Municipal Operations Platform

Version: 1.0

Document Type: Database Architecture & Schema Design

Database Engine:
PostgreSQL 17

Supporting Components:

Redis
Qdrant
Object Storage (S3/MinIO)
PostGIS

---

1. DATABASE DESIGN OBJECTIVES

The database architecture must support:

• 1,000,000+ Citizens

• 100,000+ Complaints Daily

• Thousands of Government Employees

• Real-Time Complaint Tracking

• AI-Powered Complaint Analysis

• GIS-Based Infrastructure Intelligence

• Historical Analytics

• Predictive Maintenance

• Enterprise Audit Logging

• Multi-City Deployment

---

2. DATABASE ARCHITECTURE OVERVIEW

Primary Database

PostgreSQL

Responsibilities:

User Data

Complaints

Departments

Assignments

Notifications

Audit Logs

AI Results

Analytics

Cache Layer

Redis

Responsibilities:

Sessions

OTP Storage

Rate Limiting

Live Tracking

Frequently Accessed Data

Notification Queues

Vector Database

Qdrant

Responsibilities:

Duplicate Complaint Detection

Semantic Search

Embedding Storage

Image Similarity Search

GIS Database

PostGIS Extension

Responsibilities:

Spatial Queries

Heatmaps

Complaint Clusters

Department Zones

Infrastructure Intelligence

File Storage

S3 / MinIO

Responsibilities:

Images

Videos

Voice Recordings

Resolution Proof Files

Documents

---

3. DATABASE DESIGN PRINCIPLES

UUID Primary Keys

Soft Deletes

Auditability

Horizontal Scalability

Event Logging

Partitioning

Optimized Indexing

Immutable History

---

4. ENTITY RELATIONSHIP OVERVIEW

Users

↓

Complaints

↓

Assignments

↓

Departments

↓

Status Logs

↓

Notifications

↓

AI Analysis

↓

Feedback

↓

Audit Logs

---

5. USERS TABLE

Table:
users

Purpose:
Store all platform users.

Columns:

id UUID PK

role_id UUID FK

first_name VARCHAR(100)

last_name VARCHAR(100)

email VARCHAR(255)

phone VARCHAR(20)

password_hash TEXT

profile_photo_url TEXT

preferred_language VARCHAR(20)

is_active BOOLEAN

email_verified BOOLEAN

phone_verified BOOLEAN

last_login_at TIMESTAMP

created_at TIMESTAMP

updated_at TIMESTAMP

deleted_at TIMESTAMP

Indexes:

email UNIQUE

phone UNIQUE

role_id

created_at

---

6. ROLES TABLE

Table:
roles

Columns:

id UUID PK

name VARCHAR(100)

description TEXT

created_at TIMESTAMP

Roles:

Citizen

Field Worker

Department Manager

City Administrator

Super Admin

---

7. PERMISSIONS TABLE

Table:

permissions

Columns:

id UUID PK

permission_name VARCHAR(255)

description TEXT

---

8. ROLE PERMISSIONS

Table:
role_permissions

Columns:

role_id UUID FK

permission_id UUID FK

---

9. CITIES TABLE

Table:
cities

Purpose:
Multi-city deployment.

Columns:

id UUID PK

city_name VARCHAR(255)

state_name VARCHAR(255)

country_name VARCHAR(255)

population BIGINT

created_at TIMESTAMP

---

10. DEPARTMENTS TABLE

Table:

departments

Columns:

id UUID PK

city_id UUID FK

department_name VARCHAR(255)

description TEXT

email VARCHAR(255)

phone VARCHAR(30)

is_active BOOLEAN

created_at TIMESTAMP

---

11. DEPARTMENT EMPLOYEES

Table:
department_employees

Columns:

id UUID PK

user_id UUID FK

department_id UUID FK

employee_code VARCHAR(100)

designation VARCHAR(100)

joined_at TIMESTAMP

---

12. COMPLAINTS TABLE

Table:
complaints

Purpose:
Core complaint record.

Columns:

id UUID PK

complaint_number VARCHAR(50)

citizen_id UUID FK

department_id UUID FK

master_complaint_id UUID FK

title VARCHAR(255)

description TEXT

category VARCHAR(100)

status VARCHAR(50)

priority VARCHAR(20)

severity_score DECIMAL(5,2)

fraud_score DECIMAL(5,2)

duplicate_score DECIMAL(5,2)

latitude DECIMAL(10,7)

longitude DECIMAL(10,7)

location GEOGRAPHY(Point)

address TEXT

is_duplicate BOOLEAN

is_master_complaint BOOLEAN

resolved_at TIMESTAMP

closed_at TIMESTAMP

created_at TIMESTAMP

updated_at TIMESTAMP

deleted_at TIMESTAMP

Indexes:

complaint_number UNIQUE

status

priority

category

department_id

created_at

location (GIS)

---

13. COMPLAINT IMAGES

Table:
complaint_images

Columns:

id UUID PK

complaint_id UUID FK

file_url TEXT

thumbnail_url TEXT

ai_processed BOOLEAN

created_at TIMESTAMP

---

14. COMPLAINT VIDEOS

Table:
complaint_videos

Columns:

id UUID PK

complaint_id UUID FK

file_url TEXT

duration_seconds INTEGER

created_at TIMESTAMP

---

15. COMPLAINT VOICE RECORDINGS

Table:
complaint_voice_recordings

Columns:

id UUID PK

complaint_id UUID FK

file_url TEXT

transcript TEXT

language VARCHAR(20)

created_at TIMESTAMP

---

16. COMPLAINT STATUS LOGS

Table:
complaint_status_logs

Purpose:
Immutable complaint timeline.

Columns:

id UUID PK

complaint_id UUID FK

old_status VARCHAR(50)

new_status VARCHAR(50)

changed_by UUID FK

remarks TEXT

created_at TIMESTAMP

---

17. COMPLAINT ASSIGNMENTS

Table:
complaint_assignments

Columns:

id UUID PK

complaint_id UUID FK

assigned_to UUID FK

assigned_by UUID FK

assigned_at TIMESTAMP

accepted_at TIMESTAMP

completed_at TIMESTAMP

---

18. RESOLUTION REPORTS

Table:
resolution_reports

Columns:

id UUID PK

complaint_id UUID FK

worker_id UUID FK

resolution_notes TEXT

before_image_url TEXT

after_image_url TEXT

resolution_time_minutes INTEGER

created_at TIMESTAMP

---

19. CITIZIAN FEEDBACK

Table:
citizen_feedback

Columns:

id UUID PK

complaint_id UUID FK

citizen_id UUID FK

rating INTEGER

feedback TEXT

created_at TIMESTAMP

---

20. AI ANALYSIS RESULTS

Table:
ai_analysis_results

Purpose:
Store AI inference outputs.

Columns:

id UUID PK

complaint_id UUID FK

predicted_category VARCHAR(100)

predicted_department VARCHAR(255)

classification_confidence DECIMAL(5,2)

severity_score DECIMAL(5,2)

priority_prediction VARCHAR(20)

fraud_score DECIMAL(5,2)

duplicate_score DECIMAL(5,2)

created_at TIMESTAMP

---

21. AI MODEL VERSIONS

Table:
ai_model_versions

Columns:

id UUID PK

model_name VARCHAR(255)

model_version VARCHAR(100)

accuracy DECIMAL(5,2)

deployment_date TIMESTAMP

is_active BOOLEAN

---

22. DUPLICATE GROUPS

Table:
duplicate_groups

Purpose:
Master complaint grouping.

Columns:

id UUID PK

master_complaint_id UUID FK

duplicate_count INTEGER

created_at TIMESTAMP

---

23. DUPLICATE REFERENCES

Table:
duplicate_references

Columns:

id UUID PK

duplicate_group_id UUID FK

complaint_id UUID FK

similarity_score DECIMAL(5,2)

---

24. FRAUD INVESTIGATIONS

Table:
fraud_investigations

Columns:

id UUID PK

complaint_id UUID FK

fraud_score DECIMAL(5,2)

status VARCHAR(50)

investigator_id UUID FK

notes TEXT

created_at TIMESTAMP

---

25. NOTIFICATIONS

Table:
notifications

Columns:

id UUID PK

user_id UUID FK

type VARCHAR(50)

title VARCHAR(255)

message TEXT

is_read BOOLEAN

sent_at TIMESTAMP

---

26. EMAIL LOGS

Table:
email_logs

Columns:

id UUID PK

recipient_email VARCHAR(255)

subject TEXT

status VARCHAR(50)

sent_at TIMESTAMP

---

27. SMS LOGS

Table:
sms_logs

Columns:

id UUID PK

phone VARCHAR(20)

message TEXT

status VARCHAR(50)

sent_at TIMESTAMP

---

28. PUSH NOTIFICATION LOGS

Table:
push_notification_logs

Columns:

id UUID PK

user_id UUID FK

title TEXT

status VARCHAR(50)

sent_at TIMESTAMP

---

29. AUDIT LOGS

Table:
audit_logs

Purpose:
Enterprise compliance.

Columns:

id UUID PK

user_id UUID FK

entity_type VARCHAR(100)

entity_id UUID

action VARCHAR(100)

old_values JSONB

new_values JSONB

ip_address VARCHAR(100)

created_at TIMESTAMP

Indexes:

user_id

entity_type

created_at

---

30. API REQUEST LOGS

Table:
api_request_logs

Columns:

id UUID PK

user_id UUID FK

endpoint VARCHAR(255)

method VARCHAR(20)

status_code INTEGER

response_time_ms INTEGER

created_at TIMESTAMP

---

31. DEVICE FINGERPRINTS

Table:
device_fingerprints

Purpose:
Fraud prevention.

Columns:

id UUID PK

user_id UUID FK

device_hash TEXT

browser_info TEXT

ip_address TEXT

created_at TIMESTAMP

---

32. GIS ZONES

Table:
gis_zones

Columns:

id UUID PK

zone_name VARCHAR(255)

department_id UUID FK

polygon GEOGRAPHY(POLYGON)

created_at TIMESTAMP

---

33. INFRASTRUCTURE ASSETS

Table:
infrastructure_assets

Purpose:
Future predictive maintenance.

Columns:

id UUID PK

asset_type VARCHAR(100)

asset_name VARCHAR(255)

latitude DECIMAL(10,7)

longitude DECIMAL(10,7)

installation_date DATE

condition_score DECIMAL(5,2)

created_at TIMESTAMP

---

34. PREDICTIVE MAINTENANCE RECORDS

Table:
predictive_maintenance_records

Columns:

id UUID PK

asset_id UUID FK

failure_probability DECIMAL(5,2)

predicted_failure_date DATE

recommendation TEXT

created_at TIMESTAMP

---

35. SYSTEM SETTINGS

Table:
system_settings

Columns:

id UUID PK

setting_key VARCHAR(255)

setting_value JSONB

updated_at TIMESTAMP

---

36. DATABASE PARTITIONING STRATEGY

Partition Tables:

complaints

audit_logs

notifications

api_request_logs

status_logs

Method:

Monthly Partitions

Example:

complaints_2027_01

complaints_2027_02

complaints_2027_03

Benefits:

Faster Queries

Easy Archival

Better Scaling

---

37. REDIS DATA STRUCTURES

OTP Cache

Session Cache

Rate Limits

Active Tracking

Live Complaint Updates

Notification Queue

Hot Dashboard Data

---

38. QDRANT COLLECTIONS

Collection:

complaint_embeddings

Stores:

Complaint Vector

Complaint ID

Category

Location

Timestamp

Collection:

image_embeddings

Stores:

Image Similarity Vectors

Duplicate Detection Data

---

39. BACKUP STRATEGY

Full Backup

Daily

Incremental Backup

Every Hour

Snapshot Backup

Every 15 Minutes

Geo Replication

Enabled

---

40. SCALING TARGETS

Users:

1,000,000+

Complaints:

100,000 Daily

Images:

10,000,000+

Notifications:

50,000,000+

Audit Events:

500,000,000+

Storage:

Petabyte Ready

END OF DATABASE DESIGN DOCUMENT

API CONTROL DOCUMENT

AI-Powered Civic Intelligence & Municipal Operations Platform

Version: 1.0

Document Type: API Architecture & Endpoint Specification

API Style:
REST + WebSocket

Primary Framework:
FastAPI

API Version:
v1

Base URL:

/api/v1

Authentication:

JWT Access Token

JWT Refresh Token

Role Based Access Control (RBAC)

OAuth2 Compatible

---

1. API DESIGN PRINCIPLES

Design Goals:

• Stateless APIs

• Versioned APIs

• RESTful Architecture

• OpenAPI Compatible

• High Scalability

• AI Service Integration

• Consistent Responses

• Secure By Default

---

2. STANDARD RESPONSE FORMAT

Success Response

```
{
"success": true,
"message": "Operation completed",
"data": {}
}
```

Error Response

```
{
"success": false,
"message": "Validation failed",
"error_code": "VALIDATION_ERROR",
"errors": []
}
```

---

3. AUTHENTICATION FLOW

Citizen Login

↓

Access Token

↓

Refresh Token

↓

Protected API Access

↓

Token Refresh

↓

Session Continuation

---

4. AUTH MODULE APIs

Module:

Authentication

Base Route:

/auth

AUTH-001
Register Citizen

POST

/auth/register

Purpose:

Create citizen account

Request:

```
{
"first_name": "",
"last_name": "",
"email": "",
"phone": "",
"password": ""
}
```

Response:

```
{
"user_id": "",
"status": "pending_verification"
}
```

AUTH-002
Verify OTP

POST

/auth/verify-otp

AUTH-003
Login

POST

/auth/login

Response:

```
{
"access_token": "",
"refresh_token": ""
}
```

AUTH-004
Refresh Token

POST

/auth/refresh

AUTH-005
Logout

POST

/auth/logout

AUTH-006
Forgot Password

POST

/auth/forgot-password

AUTH-007
Reset Password

POST

/auth/reset-password

---

5. USER PROFILE APIs

Base Route

/users

USER-001
Get Profile

GET

/users/me

USER-002
Update Profile

PUT

/users/me

USER-003
Upload Profile Image

POST

/users/profile-image

USER-004
Deactivate Account

DELETE

/users/me

---

6. COMPLAINT MANAGEMENT APIs

Base Route

/complaints

COMP-001
Create Complaint

POST

/complaints

Request

```
{
"title": "",
"description": "",
"latitude": "",
"longitude": ""
}
```

Response

```
{
"complaint_id": "",
"status": "submitted"
}
```

COMP-002
Upload Complaint Images

POST

/complaints/{id}/images

Multipart Upload

COMP-003
Upload Complaint Video

POST

/complaints/{id}/video

COMP-004
Upload Voice Complaint

POST

/complaints/{id}/voice

COMP-005
Get Complaint Details

GET

/complaints/{id}

COMP-006
Get Complaint Timeline

GET

/complaints/{id}/timeline

COMP-007
Citizen Complaint List

GET

/complaints/my

COMP-008
Search Complaints

GET

/complaints/search

COMP-009
Update Complaint

PUT

/complaints/{id}

COMP-010
Escalate Complaint

POST

/complaints/{id}/escalate

COMP-011
Close Complaint

POST

/complaints/{id}/close

---

7. COMPLAINT WORKFLOW APIs

Base Route

/workflow

WORK-001
Assign Complaint

POST

/workflow/assign

WORK-002
Accept Assignment

POST

/workflow/accept

WORK-003
Reject Assignment

POST

/workflow/reject

WORK-004
Update Status

POST

/workflow/status

Allowed Statuses

Submitted

Under Review

Assigned

In Progress

Resolved

Closed

Rejected

---

8. FIELD WORKER APIs

Base Route

/worker

FIELD-001
Assigned Complaints

GET

/worker/tasks

FIELD-002
Task Details

GET

/worker/tasks/{id}

FIELD-003
Submit Resolution

POST

/worker/tasks/{id}/resolve

Request

```
{
"notes": ""
}
```

FIELD-004
Upload Resolution Evidence

POST

/worker/tasks/{id}/evidence

---

9. DEPARTMENT APIs

Base Route

/departments

DEPT-001
Department Dashboard

GET

/departments/dashboard

DEPT-002
Department Complaints

GET

/departments/complaints

DEPT-003

Department Performance

GET

/departments/performance

---

10. AI COMPLAINT ANALYZER APIs

Base Route

/ai

AI-001

Analyze Complaint

POST

/ai/analyze

Input

Text

Image

Voice

Output

Category

Department

Priority

Severity

Fraud Score

Duplicate Score

AI-002

Classify Complaint

POST

/ai/classify

AI-003
Severity Prediction

POST

/ai/severity

AI-004
Priority Prediction

POST

/ai/priority

AI-005
Department Routing

POST

/ai/route

---

11. DUPLICATE DETECTION APIs

Base Route

/duplicates

DUP-001
Check Duplicate

POST

/duplicates/check

Response

{

"is_duplicate": true,
"master_complaint": "",
"similarity_score": 0.91
}

DUP-002
Duplicate Group Details

GET

/duplicates/group/{id}

---

12. FRAUD DETECTION APIs

Base Route

/fraud

FRAUD-001
Fraud Analysis

POST

/fraud/analyze

FRAUD-002
Fraud Investigation List

GET

/fraud/investigations

---

13. NOTIFICATION APIs

Base Route

/notifications

NOTIF-001

Get Notifications

GET

/notifications

NOTIF-002
Mark Read

POST

/notifications/read

NOTIF-003
Notification Preferences

PUT

/notifications/preferences

---

14. GIS & MAP APIs

Base Route

/maps

GIS-001
Complaint Heatmap

GET

/maps/heatmap

GIS-002
Complaint Clusters

GET

/maps/clusters

GIS-003
Department Zones

GET

/maps/zones

GIS-004
Infrastructure Hotspots

GET

/maps/hotspots

---

15. ANALYTICS APIs

Base Route

/analytics

ANALYTICS-001
City Dashboard

GET

/analytics/city

ANALYTICS-002
Department Dashboard

GET

/analytics/departments

ANALYTICS-003
Resolution Metrics

GET

/analytics/resolution

ANALYTICS-004
Infrastructure Health Index

GET

/analytics/health-index

ANALYTICS-005
Trend Analysis

GET

/analytics/trends

---

16. PREDICTIVE MAINTENANCE APIs

Base Route

/predictions

PRED-001
Failure Prediction

GET

/predictions/failures

PRED-002
Hotspot Prediction

GET

/predictions/hotspots

PRED-003
Maintenance Recommendations

GET

/predictions/recommendations

---

17. ADMIN APIs

Base Route

/admin

ADMIN-001
Manage Users

GET

/admin/users

ADMIN-002
Manage Departments

GET

/admin/departments

ADMIN-003
Manage Employees

GET

/admin/employees

ADMIN-004
Audit Logs

GET

/admin/audit-logs

ADMIN-005
System Settings

GET

/admin/settings

---

18. SUPER ADMIN APIs

Base Route

/super-admin

SUPER-001
City Management

SUPER-002
Platform Config

SUPER-003
AI Model Control

SUPER-004
Tenant Management

SUPER-005
Infrastructure Management

---

19. WEBHOOK APIs

Purpose:

External Integrations

WEBHOOK-001

Complaint Created

WEBHOOK-002

Complaint Resolved

WEBHOOK-003

Department Escalation

WEBHOOK-004

Fraud Alert

---

20. REAL-TIME WEBSOCKET APIs

Endpoint

/ws

WS-001

Complaint Status Updates

WS-002

Worker Assignment Events

WS-003

Department Notifications

WS-004

Live Dashboard Metrics

WS-005

Emergency Alerts

---

21. FILE UPLOAD STANDARDS

Supported Formats

Images

JPG

PNG

WEBP

Videos

MP4

MOV

WEBM

Voice

WAV

MP3

AAC

Maximum Sizes

Image

10 MB

Video

100 MB

Voice

20 MB

---

22. API RATE LIMITING

Anonymous Users

30 Requests / Minute

Authenticated Citizens

200 Requests / Minute

Workers

500 Requests / Minute

Administrators

1000 Requests / Minute

AI Endpoints

50 Requests / Minute

---

23. SECURITY CONTROLS

JWT Authentication

Refresh Tokens

RBAC

Input Validation

File Validation

Rate Limiting

API Key Protection

WAF Integration

Request Signing

CSRF Protection

XSS Protection

SQL Injection Protection

---

24. ERROR CODES

AUTH_001

Invalid Credentials

AUTH_002

Expired Token

COMP_001

Complaint Not Found

COMP_002

Invalid Location

AI_001

AI Processing Failed

AI_002

Model Unavailable

SYS_001

Internal Server Error

---

25. API VERSIONING STRATEGY

Current Version

v1

Future Versions

v2

v3

Backward Compatibility

12 Months

Version Header Support

Enabled

---

26. OPENAPI DOCUMENTATION

Swagger UI

Enabled

OpenAPI 3.1

Enabled

Interactive API Testing

Enabled

---

27. OBSERVABILITY

Request Logging

API Metrics

Latency Tracking

Error Monitoring

Tracing

Rate Limit Monitoring

Security Event Monitoring

---

28. PERFORMANCE TARGETS

Average API Response

< 200ms

AI API Response

< 800ms

Dashboard APIs

< 300ms

WebSocket Latency

< 100ms

END OF API CONTROL DOCUMENT

AI ENGINE DESIGN DOCUMENT

Project Name

AI-Powered Civic Intelligence & Municipal Operations Platform

Version: 1.0

Document Type:
AI Architecture & Machine Learning Design

---

1. AI SYSTEM OVERVIEW

The AI layer is the intelligence core of the Civic Platform.

Its responsibilities include:

1. Complaint Understanding
2. Complaint Classification
3. Department Routing
4. Priority Prediction
5. Severity Assessment
6. Duplicate Complaint Detection
7. Image-Based Issue Detection
8. Voice Complaint Processing
9. Fraud Detection
10. Predictive Infrastructure Intelligence
11. Domain-Restricted Civic Assistant

The AI system must be explainable, auditable, deterministic, and government-compliant.

---

2. HIGH LEVEL AI ARCHITECTURE

Citizen Input

(Text / Image / Voice)

↓

Input Validation Layer

↓

Domain Restriction Layer

↓

Multimodal Complaint Analysis Engine

↓

Complaint Intelligence Pipeline

↓

Classification Engine

↓

Severity Engine

↓

Priority Engine

↓

Department Routing Engine

↓

Duplicate Detection Engine

↓

Fraud Detection Engine

↓

Complaint Creation

↓

Municipal Workflow Engine

↓

Citizen Notifications

---

3. MULTI MODAL AI PIPELINE

Supported Inputs

A. Text

Example:

"There is a huge pothole near City Hospital."

B. Image

Citizen uploads photo.

C. Voice

Citizen records complaint.

D. Hybrid

Voice + Image

Text + Image

Text + Voice

Text + Image + GPS

---

4. AI COMPLAINT ANALYZER

Input

Complaint Data

↓

Text Extraction

↓

Image Analysis

↓

Location Analysis

↓

Context Analysis

↓

Issue Classification

↓

Severity Calculation

↓

Priority Prediction

↓

Department Assignment

↓

Duplicate Search

↓

Fraud Analysis

↓

Complaint Generation

Output

Category

Priority

Department

Severity Score

Fraud Score

Duplicate Match Score

Confidence Score

---

5. NLP ENGINE ARCHITECTURE

Purpose

Understand citizen complaints.

Example

"There is garbage piling up near the bus stand."

Expected Output

Category:
Garbage

Department:
Sanitation

Priority:

Medium

Confidence:
96%

---

6. COMPLAINT CLASSIFICATION CLASSES

Road

Garbage

Drainage

Water Supply

Electricity

Public Toilet

Traffic

Park Maintenance

Public Property

Other

Future Classes

Animal Control

Parking

Noise Pollution

Street Vendor Issues

Urban Forestry

---

7. NLP MODEL COMPARISON

DistilBERT

Pros

Fast

Small

Low cost

Cons

Lower accuracy

BERT Base

Pros

High accuracy

Mature ecosystem

Cons

Slower inference

RoBERTa

Pros

Highest classification accuracy

Better contextual understanding

Cons

Larger model

More infrastructure cost

MiniLM

Pros

Very fast

Excellent for embeddings

Cons

Slightly lower classification quality

FINAL RECOMMENDATION

Complaint Classification:

RoBERTa Fine Tuned

Reason:

Best balance of government-grade accuracy and reliability.

Embedding Generation:

MiniLM

Reason:

Fast semantic search and duplicate detection.

---

8. DATASET STRATEGY

Data Sources

Municipal Complaint Archives

311 Datasets

Open Government Portals

Smart City Projects

Synthetic Complaint Generation

Citizen Feedback Data

Expected Dataset Size

Phase 1

100,000 Complaints

Phase 2

500,000 Complaints

Phase 3

1 Million Complaints

---

9. DATA LABELING STRATEGY

Each Complaint Receives

Category

Department

Priority

Severity

Resolution Status

Fraud Label

Duplicate Group ID

Location Metadata

---

10. NLP TRAINING PIPELINE

Raw Complaint

↓

Cleaning

↓

Language Detection

↓

Translation (if required)

↓

Tokenization

↓

Feature Extraction

↓

Model Training

↓

Validation

↓

Model Registry

↓

Production Deployment

---

11. NLP EVALUATION METRICS

Precision

Target

92%+

Recall

Target

90%+

F1 Score

Target

91%+

Inference Latency

Target

< 200ms

---

12. COMPUTER VISION ENGINE

Purpose

Detect infrastructure issues from images.

Supported Classes

Potholes

Garbage

Water Leakage

Broken Streetlights

Open Manholes

Damaged Roads

Broken Benches

Damaged Public Property

Drainage Overflow

Illegal Dumping

---

13. OBJECT DETECTION MODEL COMPARISON

YOLOv8

Pros

Fast

Stable

Production proven

YOLOv11

Pros

Higher accuracy

Better detection quality

Modern architecture

EfficientNet

Pros

Excellent classification

Not ideal for object localization

Faster R-CNN

Pros

High accuracy

Cons

Slow inference

Expensive

FINAL RECOMMENDATION

YOLOv11

Reason

Best balance of speed and accuracy for city-wide deployment.

---

14. IMAGE TRAINING PIPELINE

Image Collection

↓

Annotation

↓

Data Augmentation

↓

Training

↓

Validation

↓

Model Registry

↓

Deployment

---

15. IMAGE AUGMENTATION

Brightness

Contrast

Rotation

Weather Simulation

Rain

Fog

Night Conditions

Blur

Compression

Occlusion

Reason

Improve real-world robustness.

---

16. VOICE PROCESSING ENGINE

Pipeline

Voice Input

↓

Speech To Text

↓

Text Validation

↓

Complaint Classification

↓

Complaint Creation

Recommended Technologies

OpenAI Whisper

Google Speech-to-Text

Azure Speech Service

Recommended

Whisper Large-v3

Reason

Supports multilingual complaints and high accuracy.

---

17. DUPLICATE COMPLAINT DETECTION

Goal

50 Citizens Reporting Same Pothole

↓

Single Master Complaint

↓

49 Linked Reports

---

18. DUPLICATE DETECTION PIPELINE

Complaint

↓

Embedding Generation

↓

Location Similarity

↓

Image Similarity

↓

Semantic Similarity

↓

Duplicate Decision

---

19. EMBEDDING STRATEGY

Model

MiniLM

Output

384-Dimensional Embedding

Storage

Vector Database

Similarity Metric

Cosine Similarity

---

20. VECTOR DATABASE COMPARISON

Pinecone

Pros

Managed

Expensive

Qdrant

Pros

Open Source

Fast

Geo Search Support

Excellent Filtering

---

pgvector

Pros

Inside PostgreSQL

Simple Deployment

Limited Large Scale Performance

FINAL RECOMMENDATION

Qdrant

Reason

Scalable civic search with geographic filtering.

---

21. DUPLICATE MATCHING FORMULA

Final Score

0.50 × Text Similarity

*

0.25 × GPS Similarity

*

0.25 × Image Similarity

Threshold

Duplicate if Score > 0.85

---

22. PRIORITY ENGINE

Priority Levels

Low

Medium

High

Critical

Scoring Factors

Issue Type

Location Importance

Severity

Citizen Impact

Safety Risk

Infrastructure Risk

---

23. PRIORITY SCORING FORMULA

Priority Score

(Severity × 0.4)

*

(Safety Risk × 0.3)

*

(Public Impact × 0.2)

*

(Location Importance × 0.1)

Thresholds

0-25

Low

26-50

Medium

51-75

High

76-100

Critical

---

24. FRAUD DETECTION ENGINE

Detect

Spam Complaints

Fake Images

Mass Complaint Abuse

Duplicate Accounts

Bot Activity

Features

Complaint Frequency

Device Fingerprint

Location Pattern

Account Age

Historical Behavior

Image Authenticity

Resolution Feedback Pattern

Model

XGBoost

Reason

Excellent explainability for government systems.

---

25. DOMAIN RESTRICTION AI

MOST CRITICAL COMPONENT

Purpose

Prevent AI misuse.

LAYER 1

Intent Classification

Detect

Coding

Math

Education

Entertainment

Civic

Non Civic

LAYER 2

Domain Relevance Detection

Check relevance to:

Infrastructure

Complaints

Municipal Services

Government Operations

LAYER 3

Prompt Guardrails

Blocked Topics

Programming

Homework

General Chat

Creative Writing

Medical Advice

Legal Advice

LAYER 4

Output Validation

Output must reference:

Complaints

Departments

Municipal Services

Status Tracking

Infrastructure

Otherwise rejected.

LAYER 5

Moderation Layer

Final policy enforcement.

Allowed

Complaint Support

Complaint Tracking

Department Information

Municipal Guidance

Rejected Example

"Write Python code"

Response

"I can only assist with civic complaints, municipal services, and infrastructure-related issues."

---

26. PREDICTIVE INFRASTRUCTURE AI

Future Capability

Predict:

Future Potholes

Garbage Hotspots

Drainage Failures

Water Leakage Zones

Streetlight Failures

Models

XGBoost

LightGBM

Random Forest

LSTM (for time series)

Recommended

LightGBM

Reason

Fast

Interpretable

Excellent tabular performance

---

27. AI OBSERVABILITY

Monitor

Classification Accuracy

Routing Accuracy

Duplicate Accuracy

Fraud Accuracy

Model Drift

Inference Latency

Prediction Quality

Citizen Satisfaction

---

28. MODEL GOVERNANCE

Version Control

MLflow

Model Registry

Required

Rollback Capability

Required

Audit Trail

Required

Government Compliance

Required

FINAL AI STACK

NLP Classification:
RoBERTa

Embeddings:
MiniLM

Speech:
Whisper Large-v3

Object Detection:
YOLOv11

Fraud Detection:
XGBoost

Prediction Engine:
LightGBM

Vector Database:
Qdrant

Model Tracking:
MLflow

Inference APIs:
FastAPI

This architecture is designed to support:

• 1 Million Citizens
• 100,000 Complaints Per Day
• Multi-City Deployment
• Government-Grade Reliability
• Real-Time Complaint Intelligence
• Strict Domain-Restricted AI Operations

INFRASTRUCTURE & DEVOPS DESIGN DOCUMENT

Project Name

AI-Powered Civic Intelligence & Municipal Operations Platform

Version: 1.0

Document Type:
Infrastructure Architecture + DevOps + Production Operations

---

1. DOCUMENT OBJECTIVE

This document defines the complete production infrastructure, cloud architecture, deployment model, networking design, containerization strategy, CI/CD architecture, monitoring stack, logging architecture, disaster recovery strategy, security hardening, and scaling strategy for a city-scale civic complaint management platform.

Target Capacity:

• 1,000,000+ Citizens

• 100,000 Complaints Per Day

• 10,000 Concurrent Users

• Thousands of Municipal Employees

• Multi-City Deployment

• High Availability Architecture

---

2. INFRASTRUCTURE PRINCIPLES

The platform must be:

Highly Available

Fault Tolerant

Secure

Scalable

Cloud Native

Observable

Disaster Recoverable

Cost Efficient

Government Deployable

---

3. CLOUD PROVIDER COMPARISON

AWS

Pros

Mature ecosystem

Excellent government adoption

Best managed services

Strong security tooling

Excellent Kubernetes support

Recommended

YES

Google Cloud

Pros

Strong AI ecosystem

Excellent data analytics

Recommended

Secondary Option

Azure

Pros

Government adoption

Enterprise integrations

Recommended

Alternative Option

FINAL RECOMMENDATION

AWS

Reason:

Most mature ecosystem for large-scale public-sector deployments.

---

4. ENVIRONMENT STRATEGY

Environment 1

Local Development

Purpose

Developer Workstations

Environment 2

Development

Purpose

Shared Engineering Environment

Environment 3

QA

Purpose

Testing and Validation

Environment 4

Staging

Purpose

Production Replica

Environment 5

Production

Purpose

Live Traffic

---

5. HIGH LEVEL CLOUD ARCHITECTURE

Internet

↓

CloudFront CDN

↓

AWS WAF

↓

Application Load Balancer

↓

Kubernetes Cluster (EKS)

↓

Microservices Layer

↓

Databases

↓

Object Storage

↓

Monitoring Systems

---

6. NETWORK ARCHITECTURE

AWS Region

Primary Region

Multi Availability Zone Setup

AZ-A

AZ-B

AZ-C

VPC

CIDR Example

10.0.0.0/16

Subnets

Public Subnet

Private Application Subnet

Private Database Subnet

Monitoring Subnet

Management Subnet

---

7. KUBERNETES ARCHITECTURE

Platform

Amazon EKS

Reason

Managed Kubernetes

High availability

Auto upgrades

Security integration

Namespaces

frontend

backend

ai-services

worker-services

monitoring

logging

security

system

---

8. CONTAINER STRATEGY

Every service runs inside Docker.

Containers

Frontend Container

API Gateway Container

Auth Service Container

Complaint Service Container

AI Service Container

Notification Service Container

Analytics Service Container

Worker Container

Monitoring Container

Logging Container

---

9. MICROSERVICES DEPLOYMENT

Service 1

Authentication Service

Responsibilities

Login

JWT

OAuth

RBAC

Service 2

Citizen Service

Responsibilities

Profiles

Preferences

Citizen Dashboard

Service 3

Complaint Service

Responsibilities

Complaint CRUD

Tracking

History

Escalation

Service 4

AI Intelligence Service

Responsibilities

Classification

Severity Prediction

Routing

Fraud Detection

Duplicate Detection

Service 5

Media Service

Responsibilities

Image Uploads

Video Uploads

Storage

Compression

Service 6

Notification Service

Responsibilities

SMS

Email

Push Notifications

Service 7

Analytics Service

Responsibilities

Heatmaps

KPIs

Reporting

Predictions

---

10. DATABASE INFRASTRUCTURE

Primary Database

PostgreSQL

Deployment

Amazon RDS PostgreSQL

Features

Multi-AZ

Automated Backups

Read Replicas

Encryption

Point-in-Time Recovery

---

11. VECTOR DATABASE INFRASTRUCTURE

Technology

Qdrant

Purpose

Duplicate Complaint Detection

Semantic Search

Complaint Similarity

Deployment

Dedicated Kubernetes Cluster

Persistent Volumes Enabled

---

12. REDIS INFRASTRUCTURE

Technology

Redis

Deployment

Amazon ElastiCache

Uses

Caching

Rate Limiting

Session Storage

Temporary Data

AI Cache

---

13. MESSAGE QUEUE ARCHITECTURE

Technology

RabbitMQ

Purpose

Background Processing

Queues

Complaint Processing Queue

Image Analysis Queue

Voice Analysis Queue

Notification Queue

Fraud Detection Queue

Prediction Queue

Analytics Queue

---

14. OBJECT STORAGE

Technology

Amazon S3

Stores

Complaint Images

Videos

Voice Files

Reports

Exports

AI Artifacts

Backups

Security

Encryption Enabled

Versioning Enabled

Private Buckets

Lifecycle Rules

---

15. CDN ARCHITECTURE

Technology

CloudFront

Purpose

Global Delivery

Benefits

Reduced Latency

Lower Backend Load

Improved User Experience

---

16. API GATEWAY LAYER

Technology

Nginx + FastAPI Gateway

Responsibilities

Authentication

Rate Limiting

Routing

Load Balancing

Request Validation

API Versioning

---

17. LOAD BALANCING

Technology

AWS Application Load Balancer

Responsibilities

Traffic Distribution

Health Checks

SSL Termination

Failover

---

18. CI/CD PIPELINE

Source Control

GitHub

Pipeline

Developer Push

↓

GitHub Actions

↓

Static Analysis

↓

Unit Tests

↓

Security Scans

↓

Docker Build

↓

Container Registry

↓

Deployment

↓

Smoke Testing

↓

Production Release

---

19. AUTOMATED QUALITY GATES

Must Pass

Unit Tests

Integration Tests

Security Scans

Container Scans

API Tests

Performance Checks

Deployment blocked if any gate fails.

---

20. INFRASTRUCTURE AS CODE

Tool

Terraform

Managed Resources

VPC

Subnets

RDS

Redis

S3

EKS

IAM

Monitoring

Networking

Security Groups

---

21. MONITORING STACK

Prometheus

Metrics Collection

Grafana

Dashboards

Metrics

CPU

Memory

Network

Disk

Database Health

Queue Health

AI Inference Latency

Complaint Processing Latency

---

22. LOGGING ARCHITECTURE

Stack

EFK

Elasticsearch

Fluent Bit

Kibana

Log Types

Application Logs

Audit Logs

Security Logs

AI Logs

Infrastructure Logs

Database Logs

---

23. ALERTING SYSTEM

Alert Channels

Email

Slack

SMS

PagerDuty

Critical Alerts

Database Down

API Down

AI Service Failure

Queue Backlog

Security Incident

High Error Rate

---

24. SECURITY ARCHITECTURE

Layer 1

AWS WAF

Protects Against

SQL Injection

XSS

Bot Attacks

DDoS

Layer 2

API Gateway Security

JWT Validation

Request Validation

Rate Limiting

Layer 3

Application Security

RBAC

OAuth

Audit Logs

Input Validation

Layer 4

Database Security

Encryption

Backups

IAM Policies

---

25. FILE SECURITY

Virus Scanning

File Type Validation

File Size Limits

Metadata Inspection

Malware Detection

Image Sanitization

Video Sanitization

---

26. SECRETS MANAGEMENT

Technology

AWS Secrets Manager

Stores

JWT Secrets

Database Credentials

API Keys

OAuth Secrets

Encryption Keys

---

27. DISASTER RECOVERY

Recovery Targets

RTO

15 Minutes

RPO

5 Minutes

Recovery Mechanisms

Database Replicas

Cross Region Backups

Infrastructure as Code

Automated Recovery

---

28. BACKUP STRATEGY

Database

Hourly Incremental

Daily Full Backup

Retention

90 Days

Files

Daily Snapshot

Cross Region Replication

Retention

180 Days

---

29. HIGH AVAILABILITY STRATEGY

Application Layer

Multiple Pods

Database Layer

Multi-AZ

Queue Layer

Clustered RabbitMQ

Cache Layer

Redis Replication

---

30. AUTO SCALING STRATEGY

Horizontal Pod Autoscaler

Metrics

CPU

Memory

Request Volume

Queue Size

AI Inference Load

Scaling Examples

Complaint Service

2 → 50 Pods

AI Service

2 → 100 Pods

Notification Service

2 → 25 Pods

---

31. COST OPTIMIZATION

Use Spot Instances

Auto Scaling

Storage Lifecycle Rules

Reserved Database Capacity

Caching Strategy

Serverless Background Tasks Where Applicable

---

32. OBSERVABILITY STRATEGY

Three Pillars

Metrics

Logs

Traces

Technology

Prometheus

Grafana

OpenTelemetry

Jaeger

Kibana

---

33. COMPLIANCE REQUIREMENTS

Government Data Policies

Audit Logging

Retention Policies

Encryption Standards

Access Controls

Data Residency Requirements

Citizen Privacy Protection

---

34. PRODUCTION READINESS CHECKLIST

✓ Security Testing Complete

✓ Load Testing Complete

✓ AI Validation Complete

✓ Backup Validation Complete

✓ Disaster Recovery Tested

✓ Monitoring Configured

✓ Alerts Configured

✓ Documentation Complete

✓ Incident Response Plan Ready

---

35. FINAL PRODUCTION TARGET

Support

1,000,000+ Citizens

100,000 Complaints Daily

10,000 Concurrent Sessions

Multiple Cities

24×7 Operations

99.95% Availability

Government-Grade Security

Real-Time AI Processing

Full Observability

Enterprise Scalability

Disaster Recovery Ready

Cloud Native Infrastructure

PRODUCTION ROADMAP & EXECUTION PLAN DOCUMENT
Project Name

AI-Powered Civic Intelligence & Municipal Operations Platform

Version: 1.0

Document Type:
Product Roadmap, Delivery Strategy, Team Structure & Execution Plan

1. DOCUMENT OBJECTIVE

This document defines the complete execution strategy for building, validating, deploying, and scaling the Civic Intelligence & Municipal Operations Platform from MVP to Multi-City Production Deployment.

The objective is to provide:

• Product Development Roadmap

• Engineering Execution Strategy

- Team Structure
- Sprint Planning
- Release Planning
- AI Development Timeline
- Infrastructure Rollout Plan
- Risk Management Strategy
- Production Deployment Strategy
- Scale Expansion Plan

2. EXECUTION PHILOSOPHY

The platform must be built using an iterative release model.

Instead of attempting to build every feature simultaneously, development should follow:

Phase 1 → Foundation

Phase 2 → Core Complaint Management

Phase 3 → AI Intelligence

Phase 4 → Municipal Operations

Phase 5 → Predictive Intelligence

Phase 6 → Multi-City Scale

3. DELIVERY MODEL

Recommended Methodology

Agile Scrum

Sprint Length

2 Weeks

Release Cycle

Every 6 Weeks

Major Production Releases

Quarterly

Architecture Reviews

Monthly

Security Reviews

Monthly

AI Model Reviews

Monthly

4. PROJECT TIMELINE

Phase 0

Planning & Architecture

Duration

4 Weeks

Phase 1

Platform Foundation

Duration

8 Weeks

Phase 2

Complaint Management System

Duration

8 Weeks

Phase 3

AI Intelligence Layer

Duration

10 Weeks

Phase 4

Municipal Operations Platform

Duration

8 Weeks

Phase 5

Predictive Infrastructure Intelligence

Duration

10 Weeks

Phase 6

Multi-City Expansion

Duration

8 Weeks

Total Initial Program

48 Weeks

Approximately 12 Months

5. TEAM STRUCTURE

Executive Layer

Product Director

Program Manager

Municipal Operations Consultant

Engineering Leadership

Principal Architect

AI Architect

Engineering Manager

DevOps Lead

Security Lead

Frontend Team

Senior React Engineers

UI/UX Designers

Mobile Engineers

QA Engineers

Backend Team

FastAPI Engineers

Database Engineers

Integration Engineers

AI Team

ML Engineers

Data Scientists

MLOps Engineers

Computer Vision Engineers

NLP Engineers

Operations Team

Cloud Engineers

DevOps Engineers

SRE Engineers

Security Engineers

6. TEAM SIZE ESTIMATION

MVP Phase

12–15 Engineers

City Scale Production

25–35 Engineers

National Scale Platform

50+ Engineers

7. PHASE 0 – DISCOVERY & ARCHITECTURE

Duration

4 Weeks

Goals

Finalize requirements

Architecture approval

Municipal workflow mapping

AI feasibility validation

Technology selection

Infrastructure planning

Security planning

Compliance review

Deliverables

PRD

Architecture Documents

Database Design

AI Design

Security Design

Roadmap Approval

8. PHASE 1 – PLATFORM FOUNDATION

Duration

8 Weeks

Objectives

Build platform core

User management

Authentication

Authorization

Infrastructure

Deployment pipelines

Features

User Registration

Login

RBAC

Departments

Roles

Profile Management

Notifications Foundation

Audit Logging

File Upload Infrastructure

Technical Deliverables

FastAPI Backend

React Frontend

PostgreSQL Setup

Redis Setup

RabbitMQ Setup

Docker Setup

Kubernetes Setup

CI/CD Pipeline

Expected Outcome

Functional platform foundation

9. PHASE 2 – COMPLAINT MANAGEMENT

Duration

8 Weeks

Objectives

Citizen complaint workflow

Municipal workflow

Tracking system

Citizen Features

Complaint Submission

Image Upload

Video Upload

GPS Location

Voice Complaint

Complaint History

Tracking Dashboard

Feedback Submission

Municipal Features

Complaint Assignment

Status Updates

Escalation

Resolution Workflow

Department Dashboard

Expected Outcome

End-to-end complaint lifecycle operational

10. PHASE 3 – AI INTELLIGENCE PLATFORM

Duration

10 Weeks

Objectives

Build AI-powered complaint processing

Modules

Complaint Classification

Severity Prediction

Priority Engine

Department Routing

Duplicate Detection

Fraud Detection

Domain Restricted Assistant

AI Deliverables

RoBERTa Model

MiniLM Embeddings

YOLOv11 Detector

Whisper Voice Engine

Qdrant Vector Search

MLflow Registry

Expected Outcome

Automated complaint intelligence

11. PHASE 4 – MUNICIPAL OPERATIONS PLATFORM

Duration

8 Weeks

Objectives

Improve government workflow efficiency

Features

Manager Dashboard

Worker Dashboard

Performance Tracking

Complaint SLA Monitoring

Workload Balancing

Infrastructure Monitoring

Escalation Management

Analytics

Department Efficiency

Complaint Trends

Resolution Performance

Citizen Satisfaction

Expected Outcome

Complete municipal operations platform

12. PHASE 5 – PREDICTIVE AI

Duration

10 Weeks

Objectives

Predict future civic issues

Prediction Models

Pothole Prediction

Garbage Hotspots

Drainage Failures

Water Leakage Risk

Streetlight Failure Prediction

Deliverables

Infrastructure Health Score

Risk Maps

Forecast Dashboards

Preventive Maintenance Insights

Expected Outcome

Transition from reactive to proactive governance

13. PHASE 6 – MULTI-CITY PLATFORM

Duration

8 Weeks

Objectives

Support multiple municipalities

Features

City Isolation

Tenant Management

City-Specific Dashboards

Regional Analytics

Cross-City Benchmarking

Shared AI Models

Expected Outcome

Commercially deployable smart-city platform

14. MVP DEFINITION

MVP Duration

16 Weeks

Includes

Authentication

Complaint Management

Citizen Dashboard

Department Dashboard

Notifications

Basic Analytics

Complaint Classification AI

Priority Engine

Department Routing

Not Included

Predictive AI

Advanced Analytics

Cross-City Features

Infrastructure Forecasting

15. VERSION RELEASE PLAN

Version 1.0

Complaint Management

Basic AI

Department Operations

Version 1.5

Duplicate Detection

Fraud Detection

Enhanced Analytics

Version 2.0

Predictive Infrastructure Intelligence

Version 3.0

Multi-City Platform

Version 4.0

National Civic Intelligence Network

16. TESTING STRATEGY

Unit Testing

Target Coverage

80%+

Integration Testing

All APIs

All Workflows

Performance Testing

100,000 Complaints Daily

Security Testing

OWASP Top 10

Penetration Testing

RBAC Validation

AI Testing

Classification Accuracy

Duplicate Detection Accuracy

Fraud Detection Accuracy

Model Drift Detection

17. PERFORMANCE TARGETS

Complaint Creation

< 2 Seconds

AI Analysis

< 3 Seconds

Dashboard Loading

< 1.5 Seconds

Notification Delivery

< 30 Seconds

API Availability

99.95%

18. RISK ANALYSIS

Risk 1

Poor AI Classification Accuracy

Mitigation

Human Review Workflow

Continuous Retraining

Risk 2

Complaint Volume Surges

Mitigation

Kubernetes Auto Scaling

Queue-Based Processing

Risk 3

Municipal Workflow Changes

Mitigation

Configurable Workflow Engine

Risk 4

Data Privacy Violations

Mitigation

Encryption

RBAC

Audit Logs

Risk 5

Fraudulent Complaints

Mitigation

Fraud Detection Engine

Verification Workflows

19. SUCCESS METRICS

Citizen Adoption Rate

Complaint Resolution Rate

Average Resolution Time

AI Classification Accuracy

Duplicate Detection Accuracy

Department Efficiency

Citizen Satisfaction Score

Infrastructure Health Index

20. YEAR 1 TARGETS

1,000,000 Registered Citizens

100,000 Complaints Per Day

90% AI Routing Accuracy

85% Duplicate Detection Accuracy

95% Department Assignment Accuracy

99.95% Platform Availability

Multi-City Deployment Capability

21. FINAL EXECUTION VISION

Year 1

Smart Complaint Management Platform

↓

Year 2

AI-Powered Municipal Operations Platform

↓

Year 3

Predictive Civic Intelligence Platform

↓

Year 5

National Smart Governance Infrastructure Network

FINAL DELIVERY SEQUENCE

Completed Documents:

✓ Product Requirement Document (PRD)

✓ Technical Architecture Document

✓ AI Architecture Document

✓ Machine Learning Architecture

✓ Backend Architecture

✓ Database Schema

✓ API Specification

✓ Security Architecture

✓ Infrastructure & DevOps Design

✓ Production Roadmap & Execution Plan

RISK ANALYSIS & GOVERNANCE FRAMEWORK DOCUMENT
Project Name

AI-Powered Civic Intelligence & Municipal Operations Platform

Version: 1.0

Document Type:
Enterprise Risk Management, AI Governance, Compliance Framework & Operational Governance

1. DOCUMENT OBJECTIVE

This document defines the governance structure, risk management framework, compliance controls, AI governance policies, operational controls, audit requirements, and accountability mechanisms required to operate a government-grade civic intelligence platform.

The platform will manage:

• Citizen Data

• Municipal Operations

• Infrastructure Intelligence

• AI-Assisted Decision Support

• Public Service Workflows

• Cross-Department Coordination

Therefore, governance must be designed as a first-class architectural component rather than an afterthought.

2. GOVERNANCE PRINCIPLES

The platform shall operate according to the following principles:

Transparency

Accountability

Auditability

Fairness

Security

Privacy Protection

Human Oversight

Data Integrity

Operational Resilience

Regulatory Compliance

3. GOVERNANCE STRUCTURE

Level 1

Platform Governance Board

Responsibilities

Strategic Oversight

Policy Approval

Risk Acceptance

Compliance Monitoring

Level 2

Municipal Operations Committee

Responsibilities

Service Quality

Complaint Resolution Policies

Department Performance Reviews

Escalation Management

Level 3

Technology Governance Committee

Responsibilities

Architecture Reviews

Security Reviews

Infrastructure Decisions

Technology Standards

Level 4

AI Governance Committee

Responsibilities

Model Approval

Bias Monitoring

Model Audits

AI Risk Reviews

Training Data Governance

Level 5

Security & Compliance Office

Responsibilities

Security Controls

Incident Management

Audit Coordination

Regulatory Compliance

4. ENTERPRISE RISK FRAMEWORK

Risk Categories

Strategic Risks

Operational Risks

Technology Risks

Security Risks

Privacy Risks

AI Risks

Infrastructure Risks

Compliance Risks

Financial Risks

Reputational Risks

5. RISK SCORING MODEL

Risk Score Formula

Risk Score

=

Likelihood × Impact

Likelihood Scale

1 Very Low

2 Low

3 Medium

4 High

5 Critical

Impact Scale

1 Negligible

2 Minor

3 Moderate

4 Major

5 Severe

Risk Categories

1–5

Low

6–10

Moderate

11–15

High

16–25

Critical

6. STRATEGIC RISKS

Risk

Low Citizen Adoption

Impact

Reduced platform effectiveness

Mitigation

Citizen awareness campaigns

Mobile-first experience

Multilingual support

Community outreach

Risk

Municipal Resistance

Impact

Poor departmental participation

Mitigation

Stakeholder workshops

Training programs

Change management initiatives

7. OPERATIONAL RISKS

Risk

Delayed Complaint Resolution

Impact

Citizen dissatisfaction

Mitigation

SLA enforcement

Escalation workflows

Manager dashboards

Risk

Incorrect Department Assignment

Impact

Workflow delays

Mitigation

AI confidence thresholds

Manual review process

Continuous model retraining

8. SECURITY RISKS

Risk

Unauthorized Access

Impact

Sensitive data exposure

Mitigation

RBAC

MFA

Zero Trust Principles

Session Monitoring

Risk

Credential Theft

Impact

Account compromise

Mitigation

MFA

Device verification

Password policies

Risk-based authentication

Risk

API Abuse

Impact

System disruption

Mitigation

Rate limiting

API gateway controls

Behavior analysis

IP reputation checks

9. PRIVACY RISKS

Risk

Citizen Data Leakage

Impact

Legal and reputational consequences

Mitigation

Encryption

Access controls

Data masking

Audit logging

Risk

Location Privacy Exposure

Impact

Citizen trust loss

Mitigation

Granular permissions

Data minimization

Location obfuscation for analytics

10. AI GOVERNANCE FRAMEWORK

Purpose

Ensure responsible use of AI throughout the platform.

Principles

Explainability

Human Oversight

Accountability

Bias Monitoring

Transparency

Auditability

Safety

11. AI DECISION BOUNDARIES

AI Can

Classify Complaints

Detect Duplicates

Estimate Priority

Suggest Departments

Predict Infrastructure Risks

Generate Municipal Insights

AI Cannot

Approve Budgets

Issue Legal Decisions

Penalize Citizens

Close Complaints Automatically

Override Human Administrators

Make Regulatory Decisions

12. HUMAN-IN-THE-LOOP POLICY

Mandatory Human Review When

AI Confidence < 85%

Fraud Risk > 75%

Critical Priority Complaints

New Complaint Categories

Model Drift Detected

Escalated Citizen Appeals

13. AI MODEL APPROVAL PROCESS

Stage 1

Data Validation

Stage 2

Training Review

Stage 3

Bias Assessment

Stage 4

Security Assessment

Stage 5

Performance Validation

Stage 6

Governance Approval

Stage 7

Production Deployment

14. AI MODEL PERFORMANCE STANDARDS

Complaint Classification Accuracy

≥ 90%

Department Routing Accuracy

≥ 95%

Duplicate Detection Precision

≥ 85%

Fraud Detection Precision

≥ 80%

Prediction Reliability

≥ 85%

15. AI BIAS MANAGEMENT

Potential Bias Sources

Location Bias

Language Bias

Department Bias

Historical Data Bias

Socioeconomic Bias

Controls

Balanced Training Data

Bias Audits

Periodic Retraining

Cross-City Validation

Human Oversight

16. DOMAIN RESTRICTION GOVERNANCE

Critical Requirement

The AI assistant is restricted exclusively to civic operations.

Allowed Topics

Complaint Creation

Complaint Tracking

Municipal Services

Infrastructure Issues

Department Information

Resolution Processes

Prohibited Topics

Programming

Mathematics

General Knowledge

Medical Advice

Legal Advice

Homework

Entertainment

Creative Writing

Career Advice

Governance Requirement

All AI responses must pass domain validation before delivery.

17. DATA GOVERNANCE FRAMEWORK

Data Categories

Public Data

Internal Data

Confidential Data

Restricted Data

Examples

Public

Infrastructure statistics

Internal

Department metrics

Confidential

Citizen information

Restricted

Security investigations

Fraud investigations

Administrative audits

18. DATA OWNERSHIP MODEL

Citizen Data

Owned by Municipality

Managed by Platform

Access Controlled

Operational Data

Owned by Municipality

AI Models

Owned by Platform Operator

Subject to Municipal Agreements

19. DATA RETENTION POLICY

Complaint Records

7 Years

Audit Logs

10 Years

AI Decision Logs

7 Years

System Logs

1 Year

Media Files

Configurable

Typically 3-5 Years

20. AUDIT FRAMEWORK

Audit Categories

Security Audit

Operational Audit

AI Audit

Financial Audit

Compliance Audit

Infrastructure Audit

Audit Frequency

Quarterly

Internal Audits

Annual

External Audits

21. AI AUDIT REQUIREMENTS

Track

Model Version

Training Dataset Version

Prediction Results

Confidence Scores

Routing Decisions

Fraud Decisions

Duplicate Detection Decisions

Every AI action must be traceable.

22. OPERATIONAL GOVERNANCE

Governed Areas

Complaint Workflows

Department Workflows

Escalation Policies

SLA Policies

Resolution Standards

Employee Accountability

23. SLA GOVERNANCE

Critical Issues

Response

Within 15 Minutes

Resolution

Within 4 Hours

High Priority

Response

1 Hour

Resolution

24 Hours

Medium Priority

Response

4 Hours

Resolution

72 Hours

Low Priority

Response

24 Hours

Resolution

7 Days

24. INCIDENT MANAGEMENT FRAMEWORK

Severity 1

Platform Outage

Major Security Breach

Severity 2

Partial Service Disruption

Severity 3

Feature-Level Issues

Severity 4

Minor Operational Issues

25. INCIDENT RESPONSE PROCESS

Detection

↓

Classification

↓

Containment

↓

Investigation

↓

Remediation

↓

Recovery

↓

Post-Mortem

↓

Governance Review

26. COMPLIANCE FRAMEWORK

Required Compliance Areas

Citizen Privacy

Data Protection

Accessibility

Government IT Policies

Cybersecurity Standards

Records Management

Digital Governance Standards

27. ACCESS GOVERNANCE

Access Principles

Least Privilege

Need-to-Know

Role-Based Access

Segregation of Duties

Periodic Access Review

28. THIRD-PARTY GOVERNANCE

All Vendors Must Meet

Security Requirements

Privacy Requirements

Availability Requirements

Audit Requirements

Compliance Requirements

Examples

SMS Providers

Email Providers

Cloud Providers

Map Services

AI Service Vendors

29. BUSINESS CONTINUITY GOVERNANCE

Objectives

Maintain Municipal Operations

Protect Citizen Services

Ensure Service Availability

Support Disaster Recovery

Recovery Governance

RTO

15 Minutes

RPO

5 Minutes

30. REPORTING FRAMEWORK

Monthly Reports

Complaint Trends

Department Performance

AI Performance

Infrastructure Health

Security Metrics

SLA Compliance

Quarterly Reports

Governance Reviews

Risk Reviews

Audit Reviews

Compliance Reviews

31. KEY GOVERNANCE KPIs

Citizen Satisfaction Score

Complaint Resolution Rate

Average Resolution Time

AI Classification Accuracy

Duplicate Detection Accuracy

Fraud Detection Precision

SLA Compliance Rate

Security Incident Rate

Audit Compliance Rate

32. GOVERNANCE MATURITY ROADMAP

Level 1

Basic Governance

Level 2

Controlled Operations

Level 3

Measured Governance

Level 4

Predictive Governance

Level 5

Autonomous Governance Intelligence

FINAL GOVERNANCE OBJECTIVE

Establish a government-grade governance framework capable of supporting:

• 1 Million+ Citizens

• 100,000 Complaints Per Day

• Multi-City Operations

• AI-Assisted Municipal Management

• Predictive Infrastructure Intelligence

• Full Auditability

• Regulatory Compliance

• Responsible AI Operations

• Transparent Public Service Delivery

DOCUMENT STATUS

Completed Documents:

✓ Product Requirement Document (PRD)

✓ Technical Architecture Document

✓ AI Engine Design Document

✓ Machine Learning Architecture

✓ Backend Architecture

✓ Database Schema Design

✓ API Specification

✓ Security Architecture

✓ Infrastructure & DevOps Design

✓ Production Roadmap & Execution Plan

✓ Risk Analysis & Governance Framework